



Autumn 2025

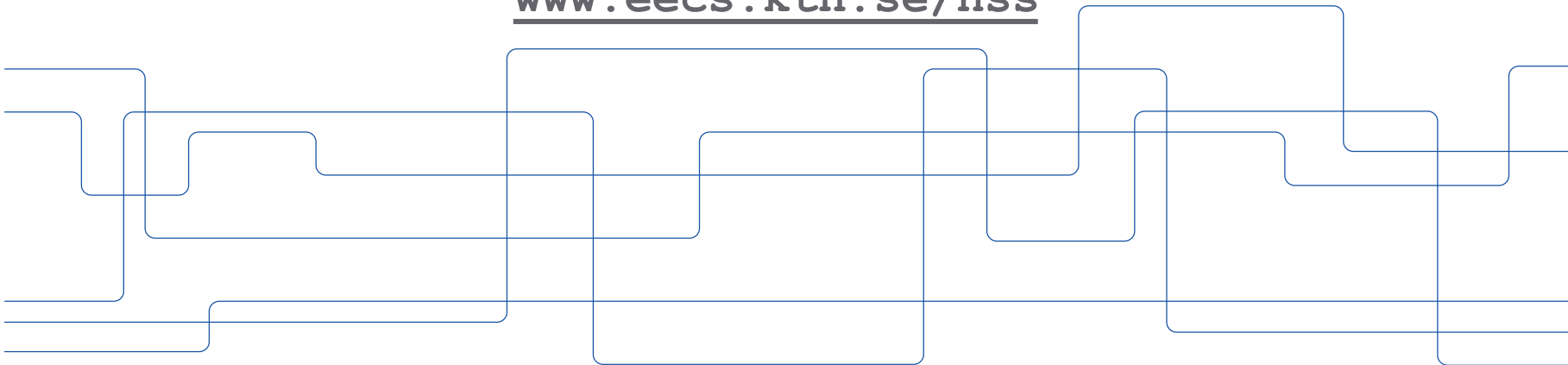
Secure localization and positioning

Discussion leads

**Andrew Roberts &
Kostas Kalogiannis**

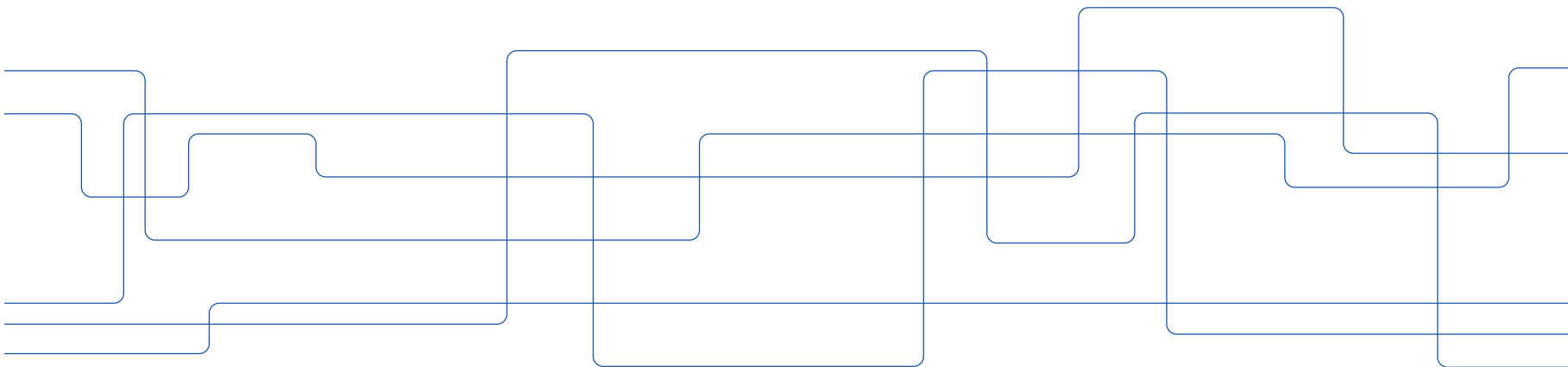
Panos Papadimitratos

www.eecs.kth.se/nss



Introduction

Secure localization and ranging





Internet of Things (IoT) applications and secure ranging/secure positioning

Direct wireless communication

Distance to other reachable devices

Multi-hop communication

Device localization and own positioning

Applications sensing the physical world

Secure Ranging

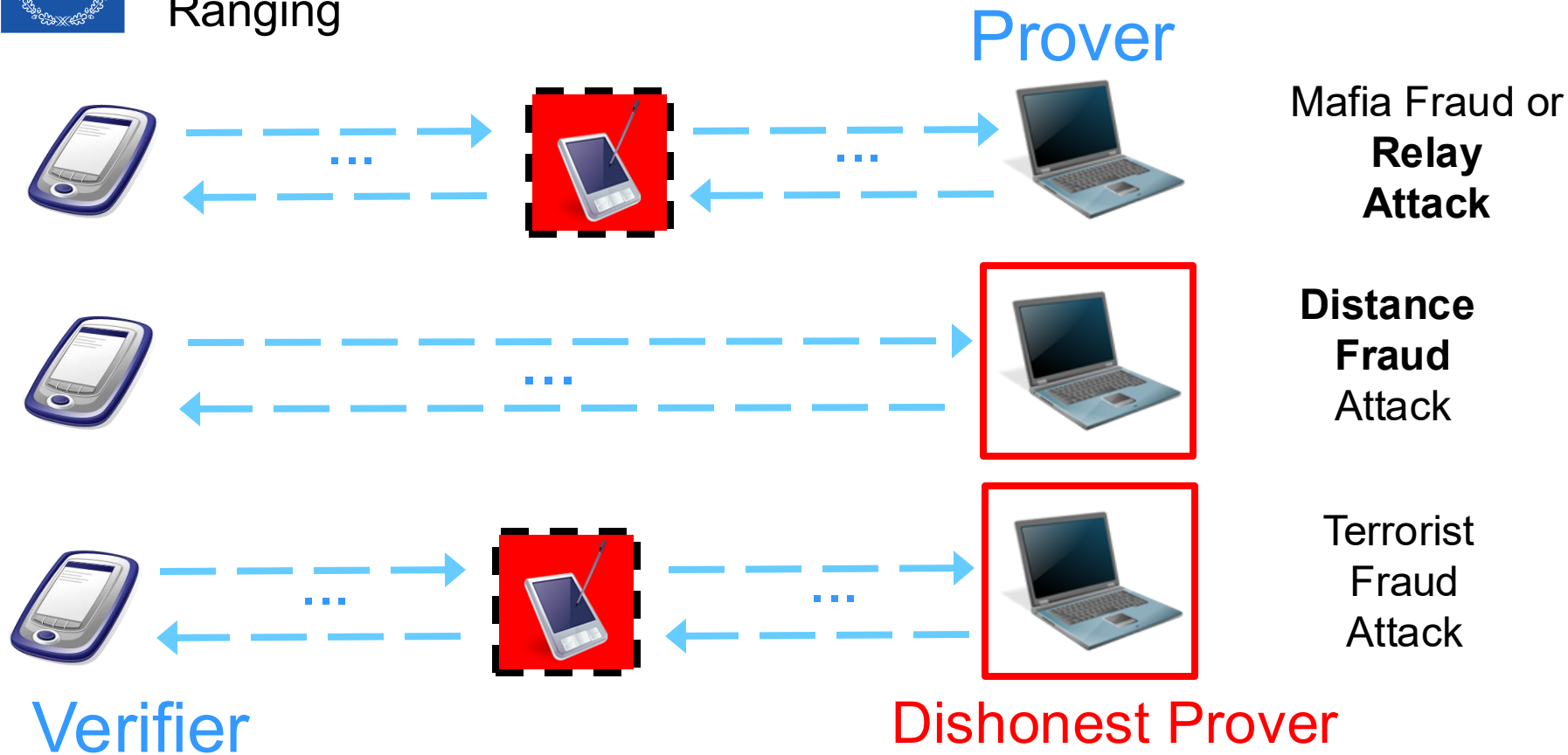
Ranging

- Measuring distances (ranges) between two parties
 - A: Obtains $d(A,B)$, an estimate of $d_{A,B}$, the actual A,B distance
 - This can be used to various extents but is also vulnerable to attacks!
- The attacker, B, provides fake inputs, to manipulate (shorten or lengthen) the $d(A,B)$ calculated by A
- Caution
 - Authentication does not solve the problem
 - Computation delays could dwarf measurements



Secure Ranging (cont'd)

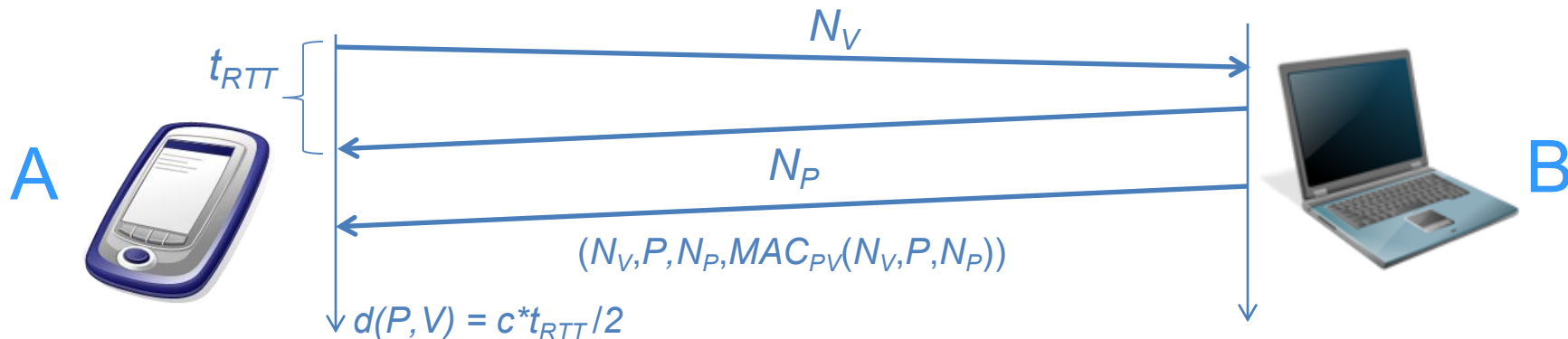
Ranging



Secure Ranging (cont'd)

Ranging

- Authenticated ranging resists external attacks (mafia frauds)
- Distance bounding resists an isolated dishonest prover (distance fraud)
- More recent protocols to defend against a colluding prover (terrorist fraud)
- Attacks by external adversaries at the physical layer
 - Undetectable by crypto alone
 - Early Detect / Late Commit: affect (reduce) the measured distance



Secure Ranging (cont'd)

Distance decreasing attacks

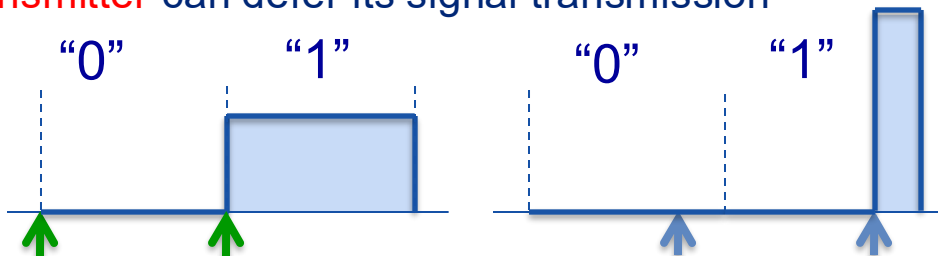
- Assume OOK:



- Early Detection* : an **adversarial receiver** does not wait for the **end of the symbol**



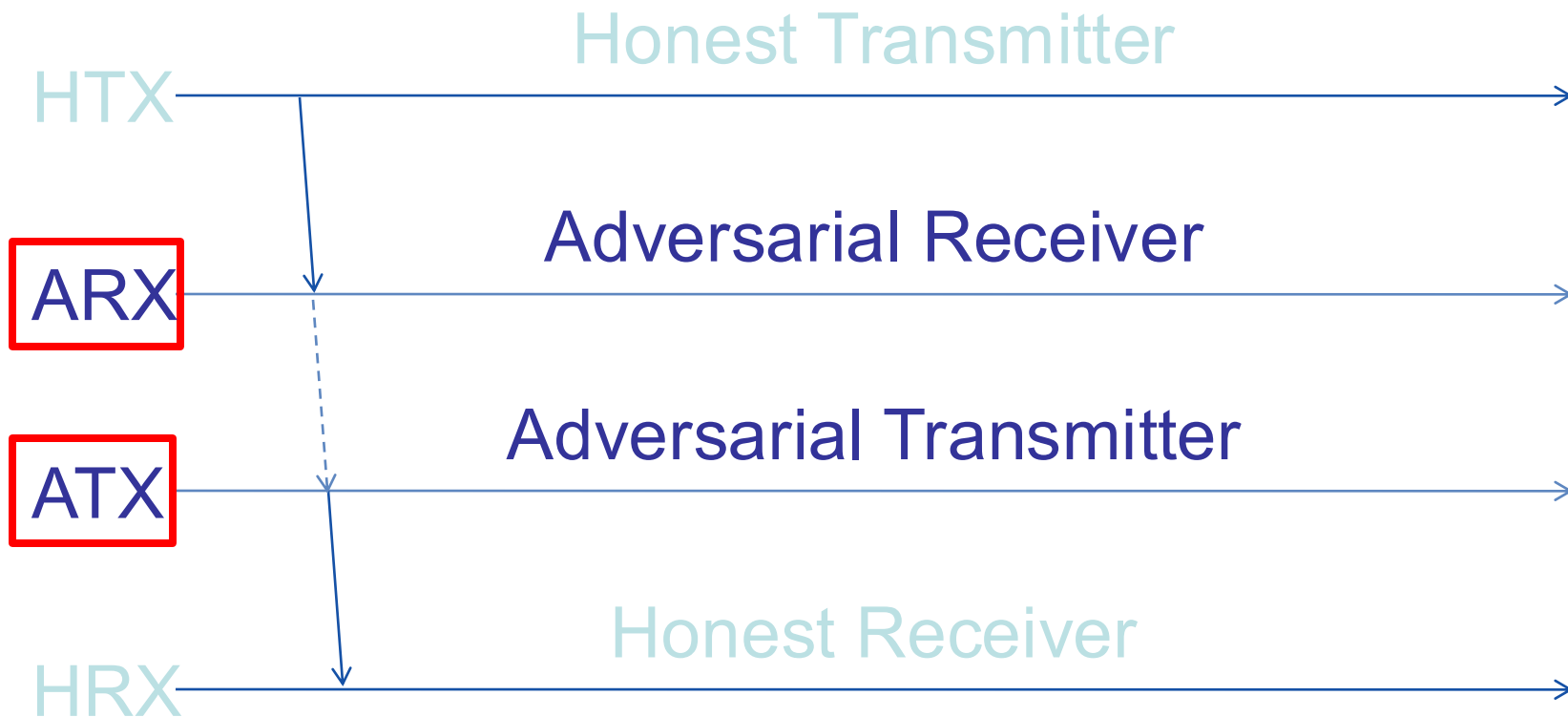
- Late Commit* : an **adversarial transmitter** can defer its signal transmission





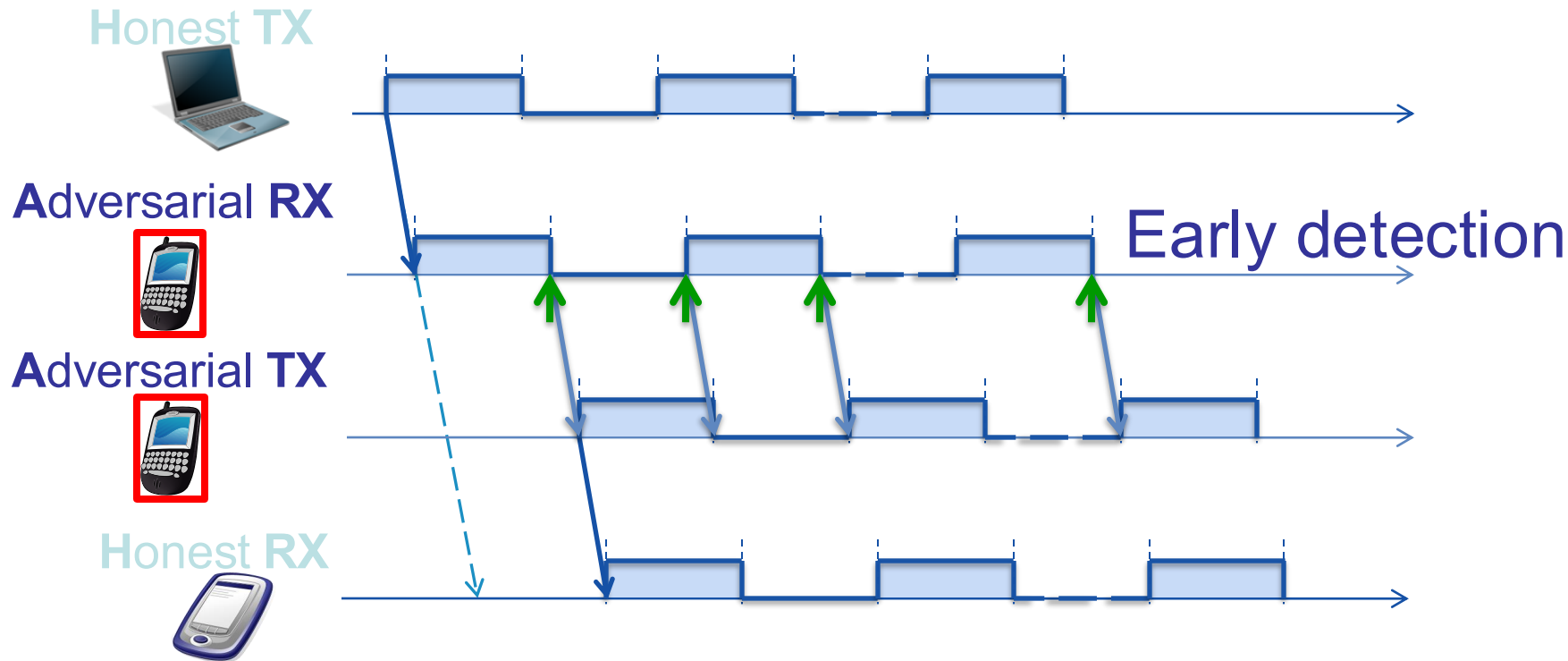
Secure Ranging (cont'd)

Distance decreasing attacks



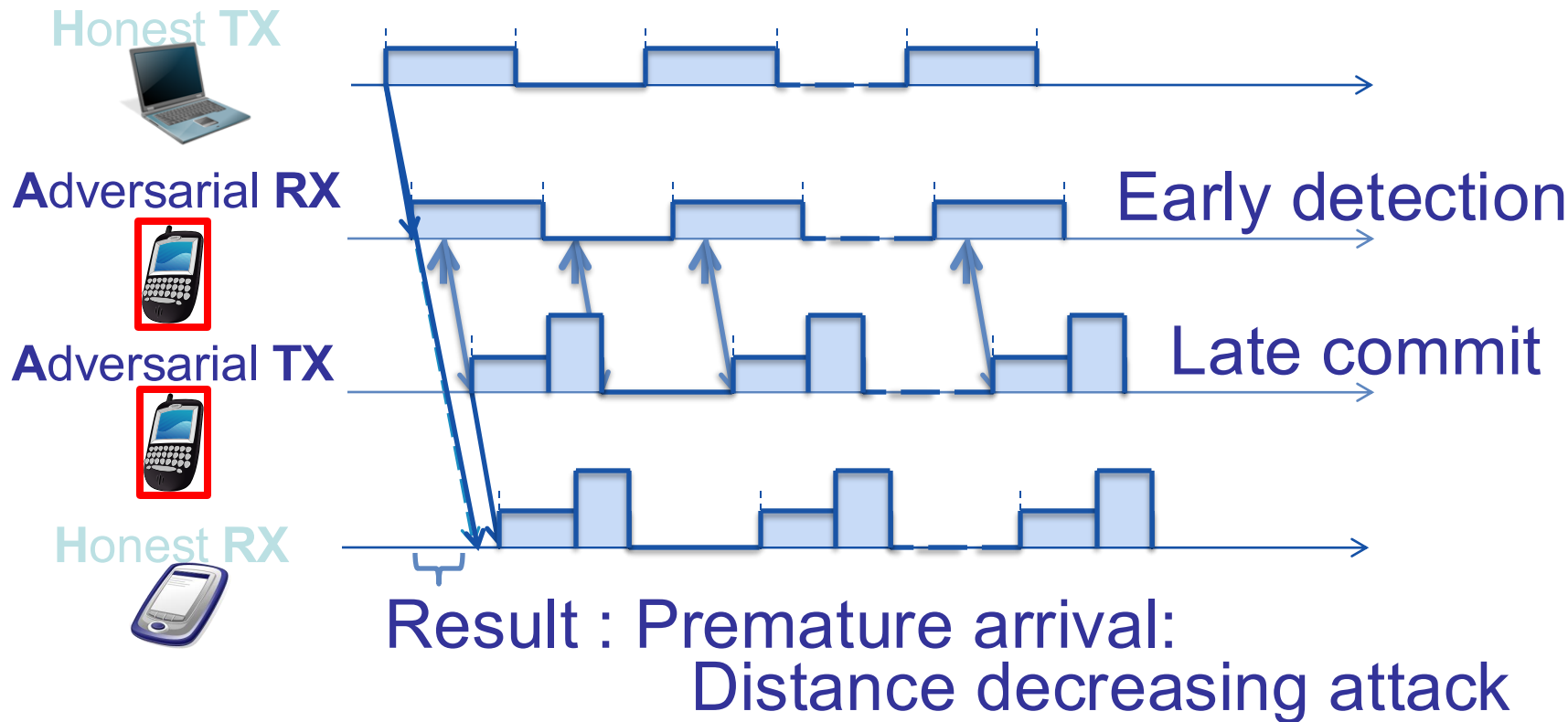
Secure Ranging (cont'd)

Distance decreasing attacks



Secure Ranging (cont'd)

Distance decreasing attacks



Secure Ranging (cont'd)

Distance decreasing attacks against UWB ranging

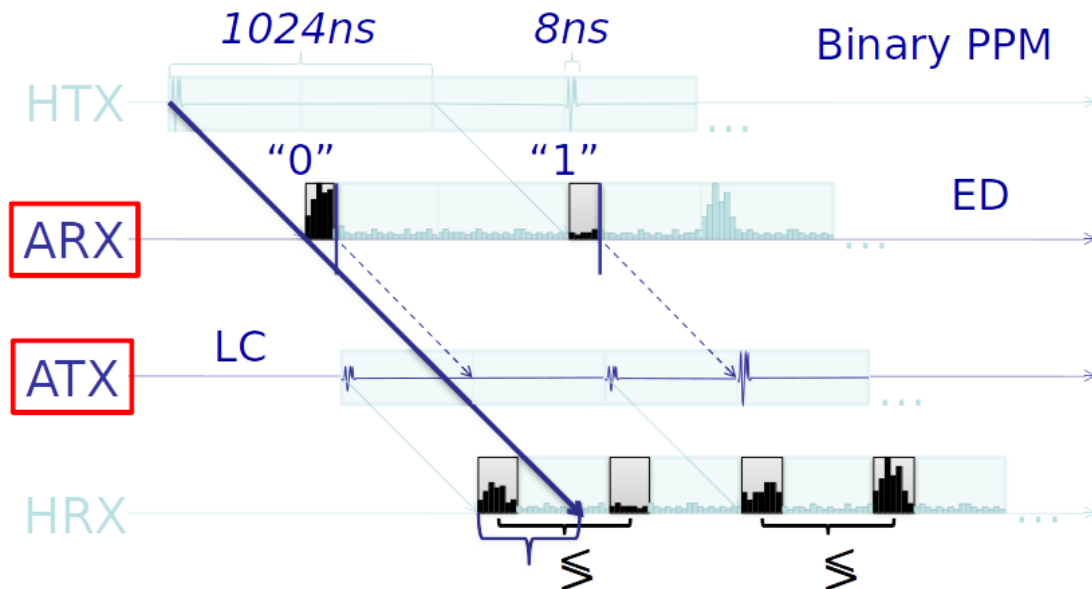
- Distance-decreasing relay attack
 - IEEE 802.15.4a standard
 - Mandatory modes
 - Energy detector receiver
 - Early detection (ED) and late commit (LC) combined
 - Distance decrease of up to 140m
 - Success rate can be made arbitrarily high

- DD attacks against GNSS

PREVIEW

M. Fleury, M. Poturalski, **P. P.**, J.-Y. Le Boudec, and J.-P. Hubaux, "Distance-decreasing Attacks Against Secure Impulse Radio Ranging," ACM WiSec 2010

+ several subsequent papers



Relay time-shift $450ns = 512ns - 62ns$
 = half symbol duration - early detection time



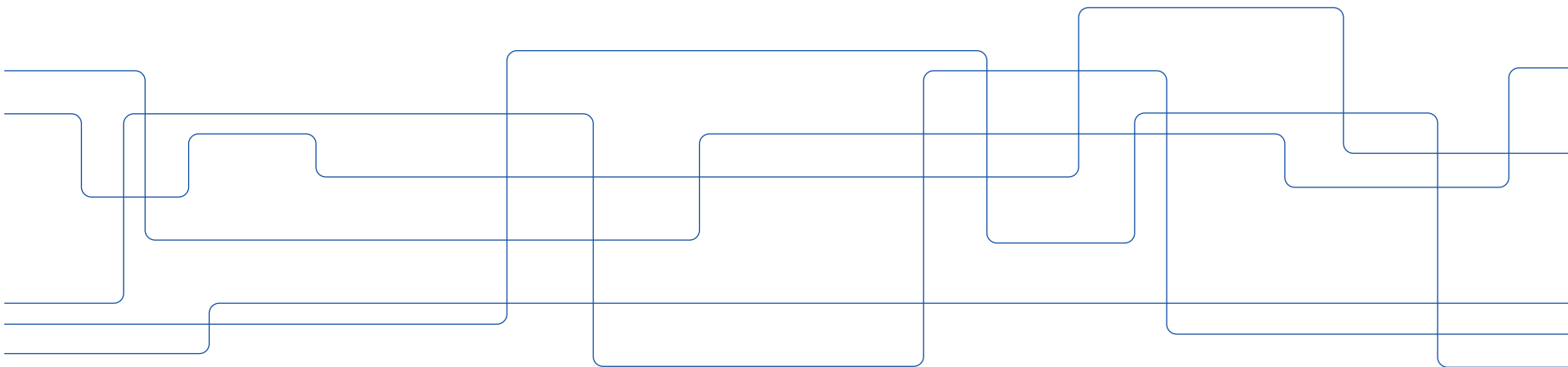
Secure Ranging (cont'd)

Reading List

- M. Poturalski, M. Flury, P. Papadimitratos, J. -P. Hubaux and J. -Y. Le Boudec, "The cicada attack: Degradation and denial of service in IR ranging," *2010 IEEE International Conference on Ultra-Wideband*, 2010, pp. 1-4, doi: 10.1109/ICUWB.2010.5616900.
- M. Poturalski, M. Flury, P. Papadimitratos, J. -P. Hubaux and J. -Y. Le Boudec, "Distance Bounding with IEEE 802.15.4a: Attacks and Countermeasures," in *IEEE Transactions on Wireless Communications*, vol. 10, no. 4, pp. 1334-1344, April 2011, doi: 10.1109/TWC.2011.020111.101219.
- M. Fiore, C. Ettore Casetti, C. -F. Chiasserini and P. Papadimitratos, "Discovery and Verification of Neighbor Positions in Mobile Ad Hoc Networks," in *IEEE Transactions on Mobile Computing*, vol. 12, no. 2, pp. 289-303, Feb. 2013, doi: 10.1109/TMC.2011.258

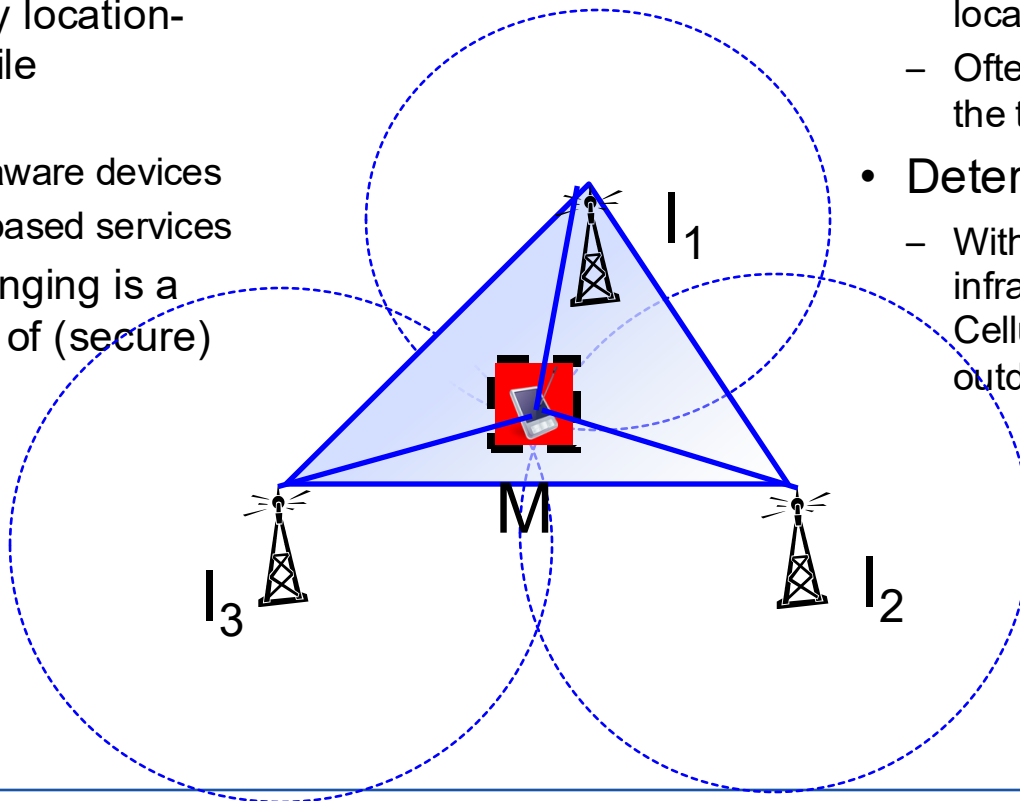
Secure localization

Obtaining a trustworthy position



Secure localization

- Increasingly location-based mobile computing
 - Location-aware devices
 - Location-based services
- (Secure) ranging is a component of (secure) localization

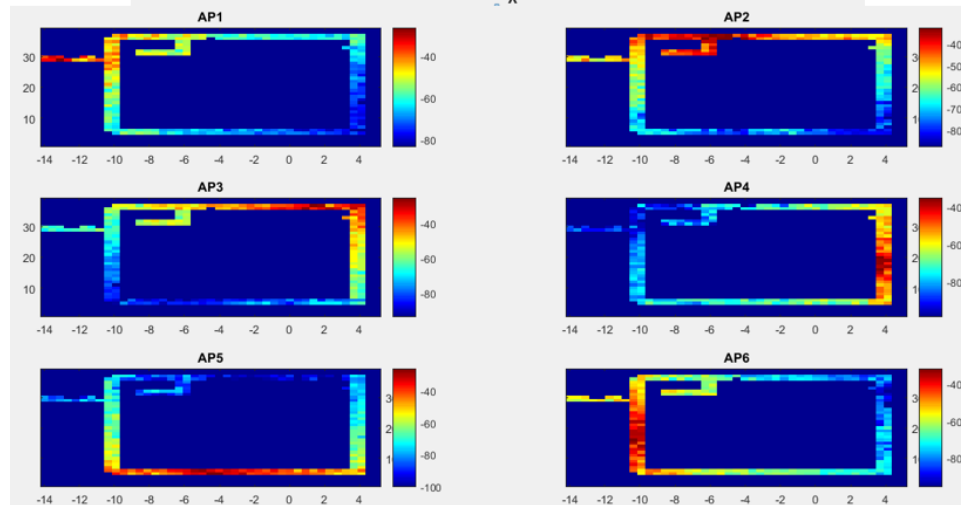
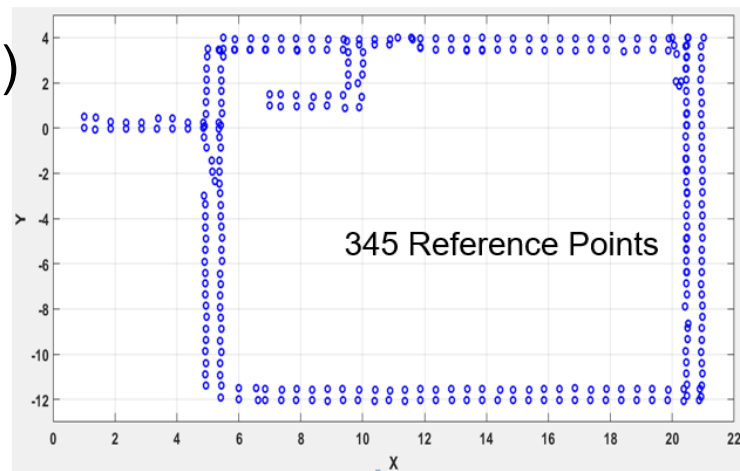


- Determine the location of a (another) device
 - Could be as simple as asking a location-aware device to report its location
 - Often, some infrastructure performs the task
- Determine own location
 - With the help of own equipment and infrastructure: WiFi (indoors), Cellular (outdoors), GNSS (global, outdoors)

Secure localization (cont'd)

Indoors

- Often uses Wi-Fi
- Recent developments on machine learning and sensor fusion
- Precise localization using consumer devices
 - Usually a combination of protocol-based ranging and physical layer properties
- Security and privacy needed



M. Hoang, X. Dong, T. Lu, B. Yuen, R. Westendorp, 2019. WiFi RSSI Indoor Localization. Available at: <https://dx.doi.org/10.21227/1yd5-rn96>.

Secure localization (cont'd)

Outdoors

- GNSS
- Signals of opportunity
 - Cellular
 - OFDM transmitters
- Communication protocols are secure (most of them), but the opportunistic usage is not!
- Several commercial products exist too, but robustness to attackers is unexplored

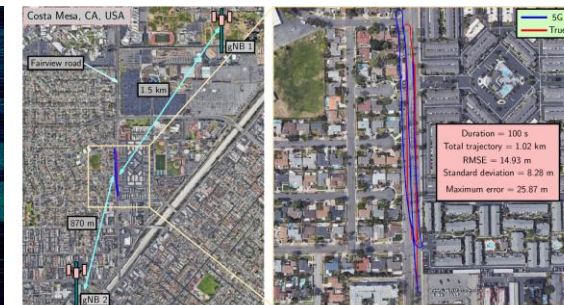
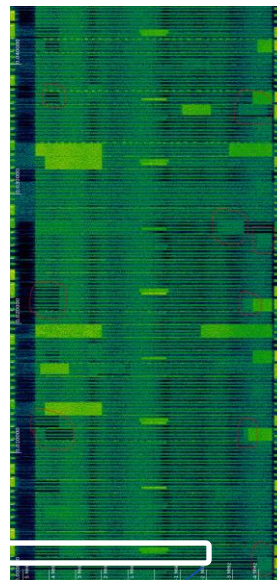
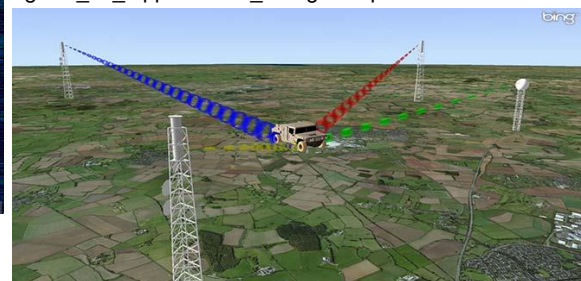
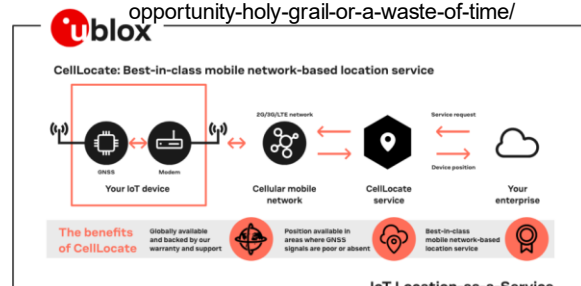
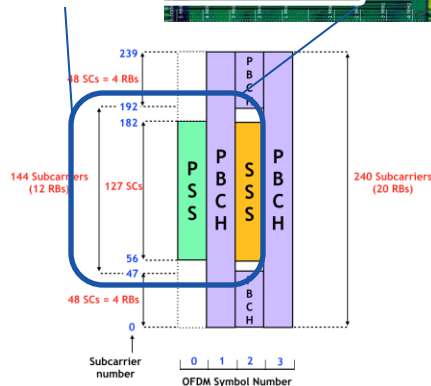


Fig. 5. Environmental layout, 5G/4G/LTE transmitters, and the measured trajectories. The 5G localization solution exhibited a superior RMSE.

http://aspin.eng.uci.edu/papers/Assessing_Real_5G_Signals_for_Opportunistic_Navigation.pdf



<https://www.gpsworld.com/signals-of-opportunity-holy-grail-or-a-waste-of-time/>



<https://www.u-blox.com/en/product/celllocate>



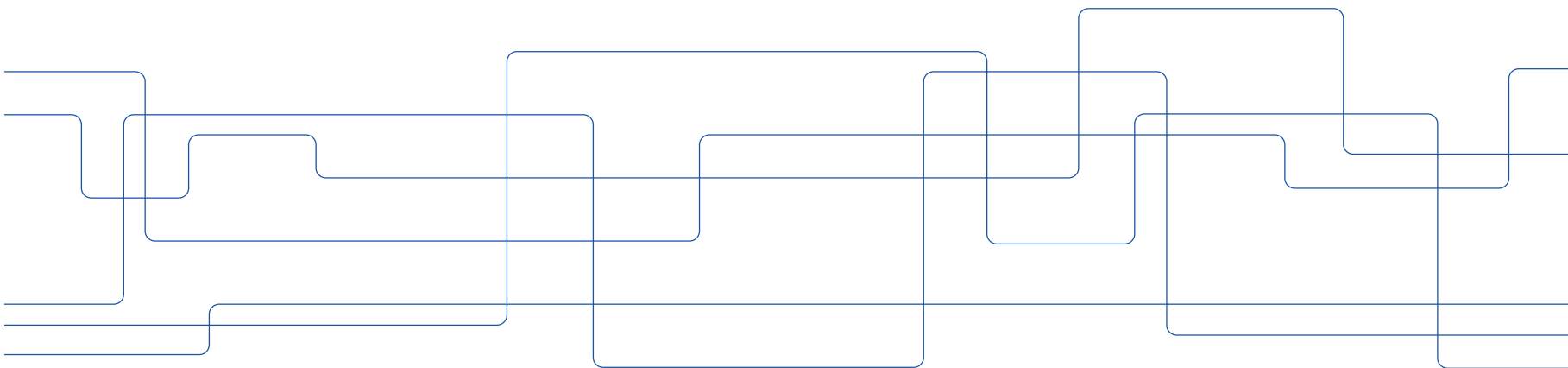
Secure localization (cont'd)

Reading List (no security)

- D. Vasisht, S. Kumar, and D. Katabi, “Decimeter-level localization with a single wifi access point,” in NSDI, 2016.
- M. Kotaru, K. Joshi, D. Bharadia, and S. Katti, “Spotfi: Decimeter level localization using wifi,” in SIGCOMM, 2015.
- K. Jiokeng, G. Jakllari, A. Tchana, and A.-L. Beylot, “When FTM discovered music: accurate wifi-based ranging in the presence of multipath,” in IEEE INFOCOM, 2020
- C. Ma, B. Wu, S. Poslad, and D. R. Selviah, “Wi-fi RTTranging performance characterization and positioning system design,” IEEE Transactions on Mobile Computing, 2020.
- J. He and H. C. So, “A Hybrid TDOA-Fingerprinting-Based Localization System for LTE Network,” in *IEEE Sensors Journal*, vol. 20, no. 22, pp. 13653-13665, 15 Nov.15, 2020, doi: 10.1109/JSEN.2020.3004179.
- L. Ni, Y. Wang, H. Tang, Z. Yin and Y. Shen, “Accurate Localization Using LTE Signaling Data,” *2017 IEEE International Conference on Computer and Information Technology (CIT)*, 2017, pp. 268-273, doi: 10.1109/CIT.2017.59.

Global navigation satellite systems (GNSS)

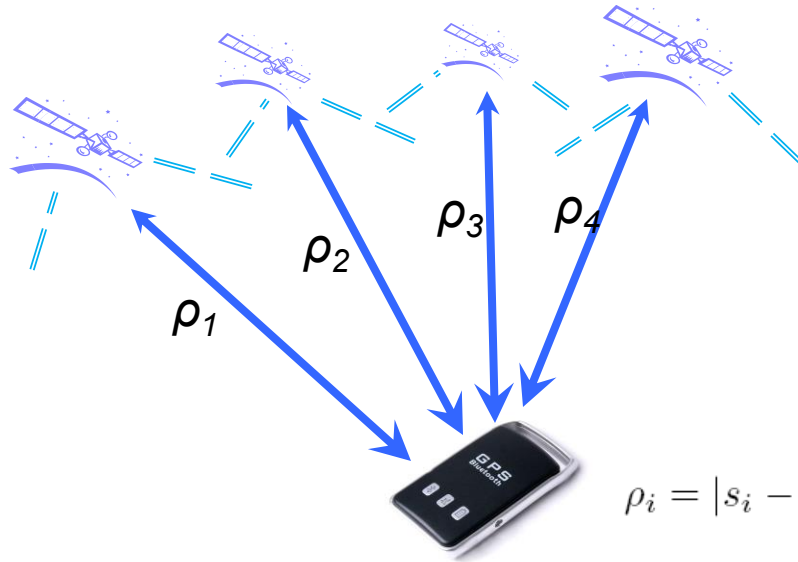
GNSS security: attacks and defenses



Secure localization (cont'd)

Global Navigation Satellite Systems (GNSS)

Localization: determining own position, typically outdoors



$$\rho_i = |s_i - loc_V| + c \cdot t_V$$

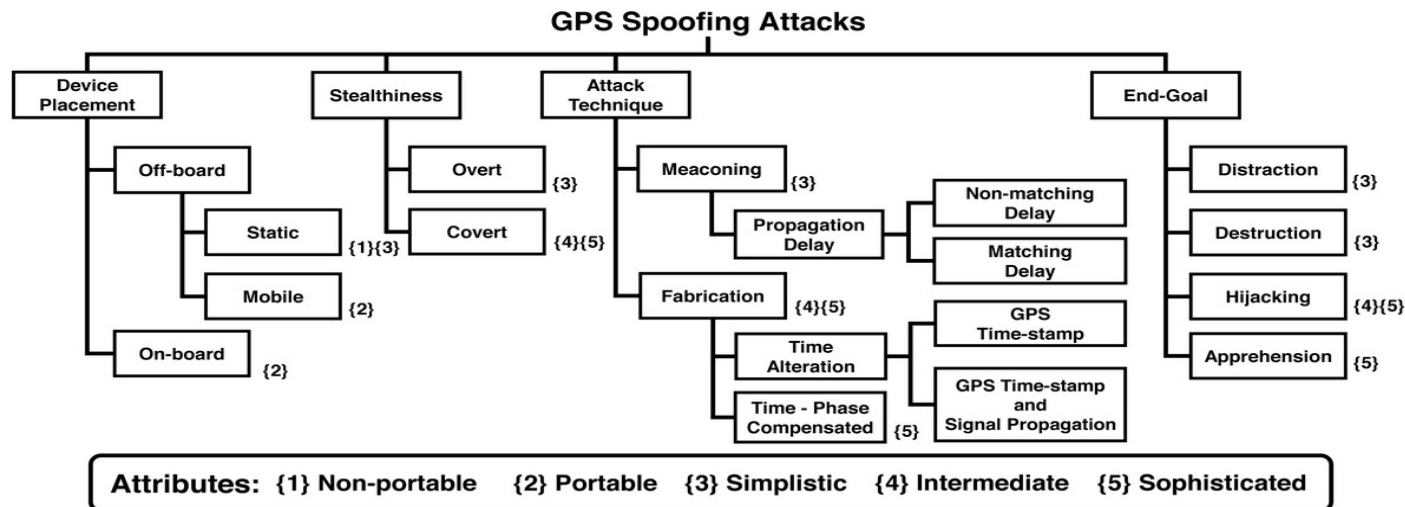
GPS receiver, V

1. Receive NAV_i from satellite S_i at position s_i
2. Estimate the NAV_i propagation delays, and thus V - S_i distances (pseudoranges), ρ_i
3. Minimize the system of equations given by the ρ_i
4. Obtain own position, loc_V , and clock correction, t_V

Secure localization (cont'd)

Attacking GNSS

- Compromise the device: hard
- Compromise the infrastructure: much harder
- Interfere with the infrastructure-to-device wireless communication: easy
 - Jam: Denial of Service
 - Spoof: Overwrite legitimate transmissions with synthesized ones: Control loc_v and t_v



doi.org/10.7717/peerj-cs.507

Attacking GNSS Localization

GNSS jamming

- During a light show, a GNSS jamming caused drones to fall out of the sky, causing public panic.



Drones

<https://www.youtube.com/watch?v=Udb0ATR8AT4>



Attacking GNSS Localization (cont'd)

GNSS jamming

- Deliberate jamming, blocking or interference with wireless communications
- Disrupt communications by decreasing the signal-to-noise ratio, disrupt information flow
- Usually distinguished from benign interference due to device malfunction or accidental circumstances
 - Unintentional jamming: equipment accidentally radiates a signal

Attacking GNSS Localization (cont'd)

GNSS jamming devices



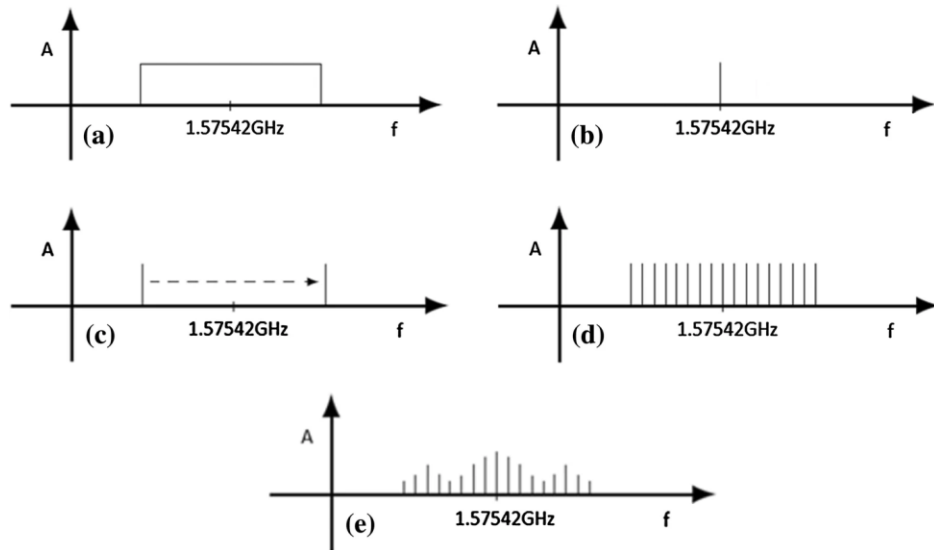
Range from \$20 to millions, commercial to military use

<https://www.vehicletrackingexperts.co.uk/gps-jammers/>
<https://trackimo.com/different-methods-of-jamming-gps/>
<https://mctech-jammers.com/solutions/military-defense.html>

Attacking GNSS Localization (cont'd)

Jamming using low-cost SDR platforms

- Barrage Jamming (a)
 - Send signals in many bands of frequencies compared to the bandwidth of any single one
- Tone Jamming (b)
 - Send of one or more CW signals specifically placed in programmed frequency values
- Sweep Jamming (c)
 - Shifting a jammer's full power from one frequency to another
- Successive Pulses Jamming (d)
 - Similar to (a), but not do it uniformly
- Protocol-Aware Jamming (e)
 - Designed for specific protocols



- Civilian maritime GNSS jamming testbed in the Baltic Sea

-
- Jammer (N) Law enforcement vessel NEUSTRELITZ
- (A) AARON
- Baltic Sea
- Hiddensee
- (B) BALTIC TAUCHER II
- N B A

Power/frequency (dB/Hz)

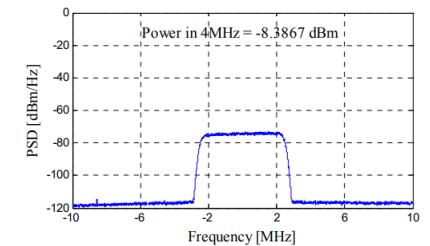
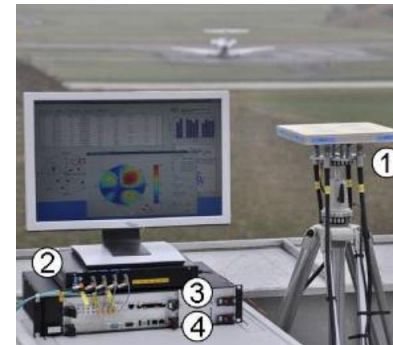
Frequency (GHz)

GPS L1
Galileo E1

Glonass G



- GNSS measurements under jamming with multi-antenna receiver
- Galileo test environment
- Hardware components
 - ① E1/E5 active antenna array with calibration network
 - ② E1/E5a front end with four channels for each frequency band
 - ③ An FPGA board with integrated analogue to digital and digital to analogue converters
 - ④ An embedded Intel 2 GHz dual core PC card



Power spectral density at the front-end output with 4 MHz AWGN interference

Attacking GNSS Localization (cont'd)

Paris observatory jamming

- November 2018: GNSS signal reception in the L1-band has been jammed by a powerful signal transmitted near to Observatoire de Paris (frequency band allocated to space to Earth satellite telecommunication)
- Paper describes the jamming signal and its consequences
 - The signal power was more than 35 dB above the GNSS spread-spectrum power level
 - All activities depending on multi-GNSS signal reception were impacted by the jamming signal



The center frequency of jamming is 1575.2 MHz. The span is 200 MHz. The GNSS allocated L1-band from 1559 MHz to 1610 MHz is in green.



Attacking GNSS Localization (cont'd)

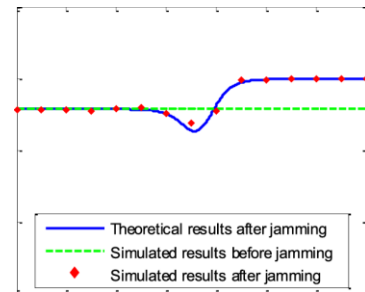
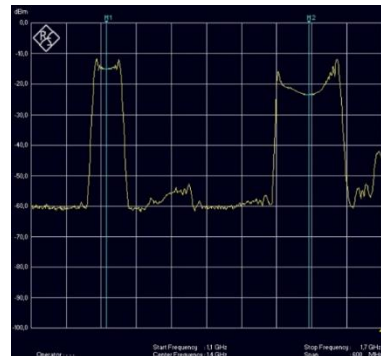
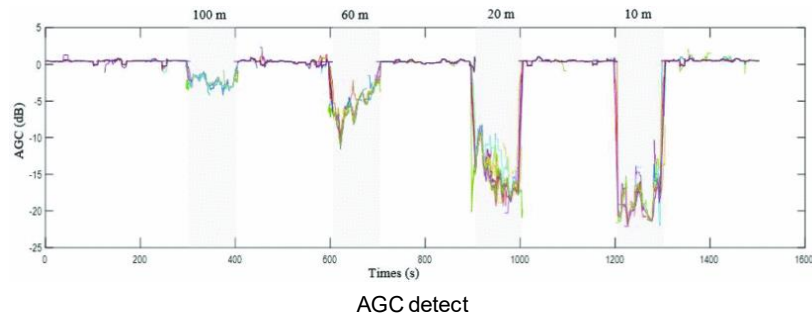
Reading list on GNSS jamming

- R. Ferreira, J. Gaspar, P. Sebastiao, and N. Souto, “Effective GPS jamming techniques for UAVs using low-cost SDR platforms,” *Wireless Personal Communications*, vol. 115, no. 4, pp. 2705–2727, 2020.
- D. Medina, C. Lass, E. P. Marcos, R. Ziebold, P. Closas and J. García, "On GNSS Jamming Threat from the Maritime Navigation Perspective," 22th International Conference on Information Fusion (FUSION), 2019, pp. 1-7, doi: 10.23919/FUSION43075.2019.9011348.
- M. Cuntz, A. Konovaltsev, M. Sgammini, C. Hattich, G. Kappen, M. Meurer, A. Hornbostel, and A. Dreher, “Field test: jamming the DLR adaptive antenna receiver,” in *Proceedings of the 24th International Technical Meeting of The Satellite Division of the Institute of Navigation (ION GNSS 2011)*, 2011, pp. 384–392.
- P. Uhrich, M. Abgrall, F. Riedel, B. Chupin, J. Achkar, and G. D. Rovera, “A powerful signal nearby I1 frequency band jamming GNSS stations in observatoire de paris,” *ITU J. ICT Discov*, vol. 2, 2019.

Defending GNSS

Energy detectors

- Test statistics
 - Correlator output power (normalized by estimated receiver noise)
 - Standard deviation of correlator output power
 - Carrier phase vacillation
 - Automatic gain control (AGC) level
- Advantage
 - Data usually can be available from a standard commercial GNSS receiver
 - Can be used to detect any type of signal
- Disadvantage
 - Hardware dependent



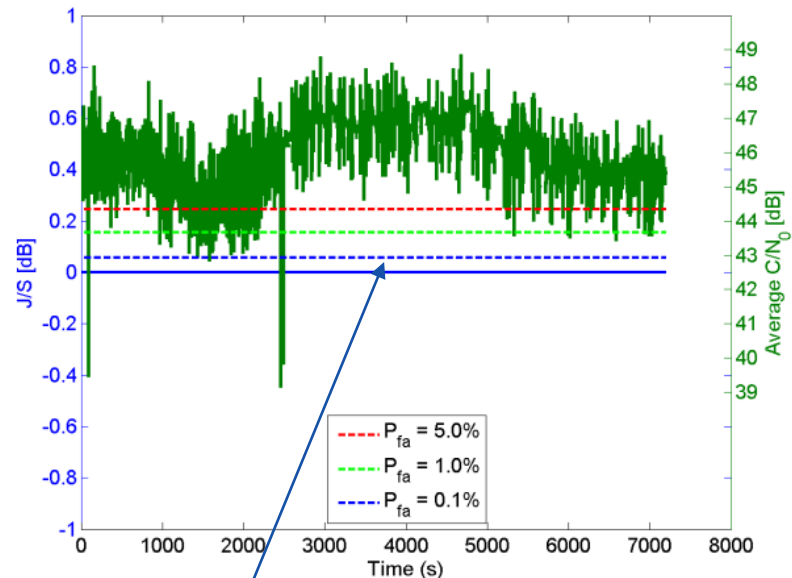
Jamming effect on correlation due to the center located jammer

Spectrum of the RX-700 jammer. The vertical blue lines show the GPS L1 and L2 center frequencies respectively

Defending GNSS (cont'd)

C/N0-based detectors

- Carrier-to-noise (C/N0) ratio is the ratio of the useful power of a certain GNSS satellite signal to the noise power
- Advantage
 - Can detect any type of interference signal
- Disadvantage
 - Cannot distinguish between decreased GNSS signal power and increased interference power
 - Only useful if the GNSS received signal strength is fairly constant



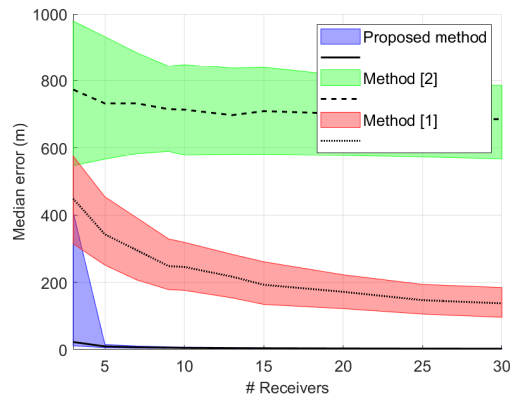
Decision thresholds and false alarm rates

Defending GNSS (cont'd)

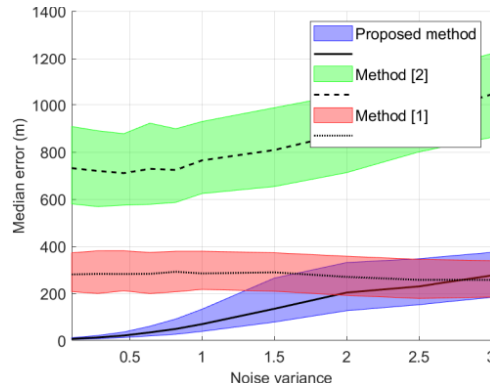
Participatory jamming detection and localization

- Embedded GNSS receivers to provide measurements for participatory jamming detection and localization.
- Jamming localization algorithm based on participatory sensing, that exploits AGC and C/N0 from commercial GNSS receivers
- Measurements are from a large ensemble of voluntary contributors
- Does not need any pre-calibration or channel knowledge

Error and 25th to 75th percentile (shaded areas) versus different number of receivers



Error and 25th to 75th percentile (shaded areas) versus different measurement noise powers



Defending GNSS (cont'd)

Navigating with signals of opportunity under jamming

- This paper presents a radio simultaneous localization and mapping (radio SLAM) approach using signals of opportunity
- The experiments took place at Edwards Air Force Base, California, USA, during which GPS was intentionally jammed with jamming-to-signal ratio as high as 90 dB
- Fuse cellular/Wi-Fi with sensors (e.g., inertial measurement unit (IMU), lidar, etc.), digital maps, and/or other signals (e.g., GNSS)

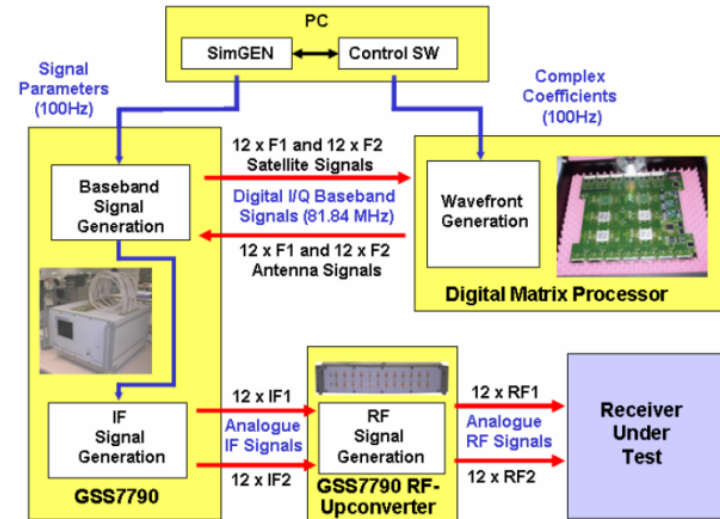


Defending GNSS (cont'd)

Array receivers for anti-jamming

- Work on the use of GNSS constellation simulators for the evaluation of advanced anti-jamming techniques of GNSS receivers using multiple antennas
- Multi-antenna GNSS constellation simulator
 - One Main Host PC
 - Two Auxiliary Host PCs
 - Two Timing Masters (TM1 and TM2)
 - Eight identical single antenna chassis signal generators (Aux 1 to Aux 8)

Simulator



Block diagram



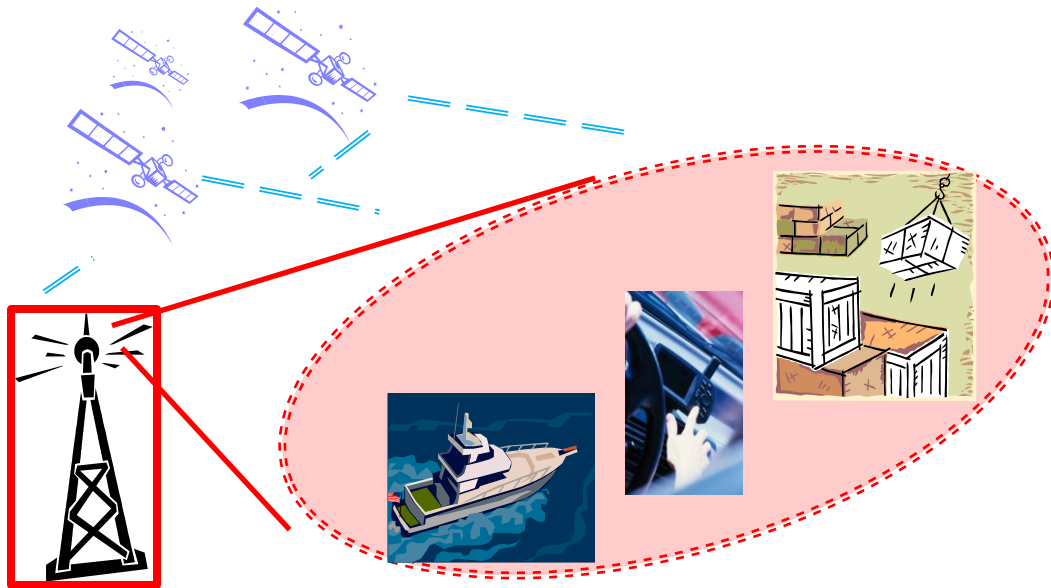
Defending GNSS (cont'd)

Reading List on anti-jamming

- E. Axell, GNSS interference detection, Feb-2014. [Online]. Available: <https://www.foi.se/rest-api/report/FOI-R--3839--SE>. [Accessed: Nov-2022].
- G. K. Olsson, E. Axell, E. G. Larsson, and P. Papadimitratos, "Participatory sensing for localization of a GNSS jammer," arXiv preprint arXiv:2204.13974, IEEE/ION Position, Location and Navigation Symposium (PLANS), Monterey, California, April 2023
- Z. M. Kassas, J. Khalife, A. A. Abdallah and C. Lee, "I Am Not Afraid of the GPS Jammer: Resilient Navigation Via Signals of Opportunity in GPS-Denied Environments," in IEEE Aerospace and Electronic Systems Magazine, vol. 37, no. 7, pp. 4-19, 1 July 2022, doi: 10.1109/MAES.2022.3154110.
- A. Konovaltsev, E. P. Marcos, M. Cuntz, M. Meurer, R. Wong, G. Buesnel and W. Lange, "Development of Array Receivers with Anti-Jamming and Anti-Spoofing Capabilities with Help of Multi-Antenna GNSS Signal Simulators," in Proceedings of the 32nd International Technical Meeting of the Satellite Division of The Institute of Navigation (ION GNSS+ 2019), Miami, Florida, September 2019.

Attacking GNSS Localization (cont'd)

Spoofing



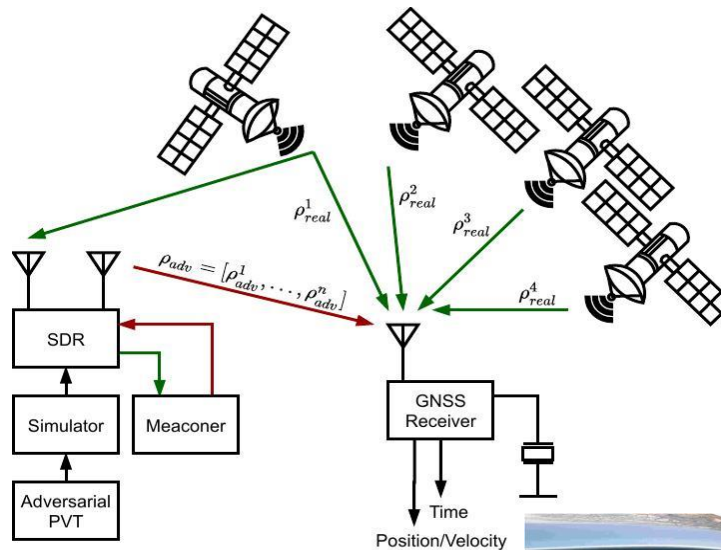
- Attacker: Record and replay, or forge, GPS signals, overwriting the legitimate GPS signals
- System: GPS receiver locks on spoofed signals
- Consequence: User is provided with a false, attacker-controlled location

Attacking GNSS Localization (cont'd)

Spoofing

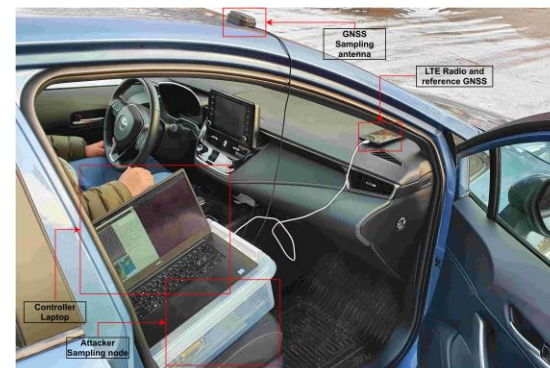
GNSS attack methods

- Simulation
 - Asynchronous zero-knowledge synthetic signals
- Intermediate Spoofer
 - Live sky simulation: synchronous or asynchronous
- Meaconing
 - Record and replay or relay attacks



<https://news.comell.edu/stories/2008/09/gps-receivers-can-be-spoofed-say-cu-researchers>

<https://doi.org/10.1109/AERO53065.2022.9843381>

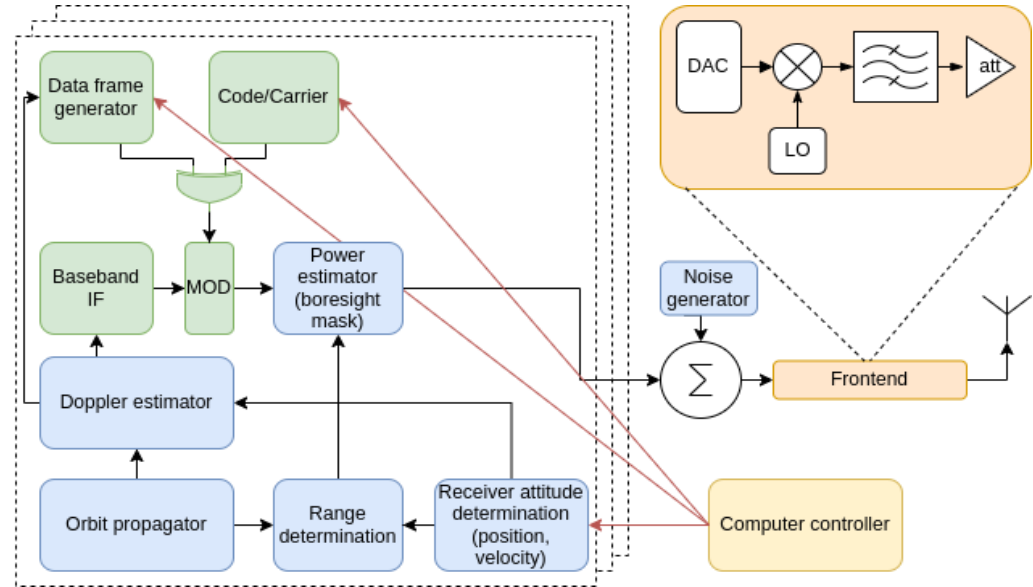


<https://www.ion.org/itm/abstracts.cfm?paperID=10822>

Attacking GNSS Localization (cont'd)

Spoofing

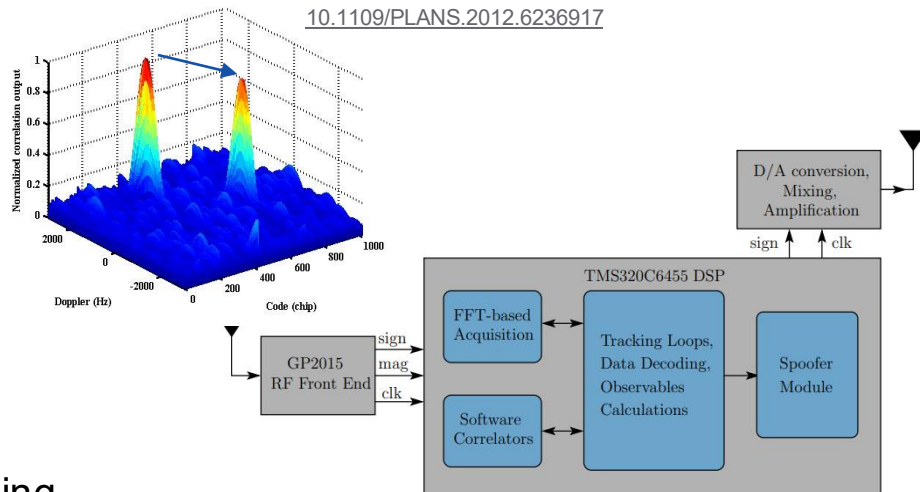
- GNSS (civilian) known structure
 - One can build own GNSS transmitter
 - <https://www.gps.gov/technical/icwg/>
 - https://www.gsc-europa.eu/sites/default/files/sites/all/files/Galileo_OS_SIS_ICD_v2.0.pdf
- Jamming to capture locked target
 - Dependency between receiver clock stability and required jamming duration



Attacking GNSS Localization (cont'd)

Intermediate spoofer

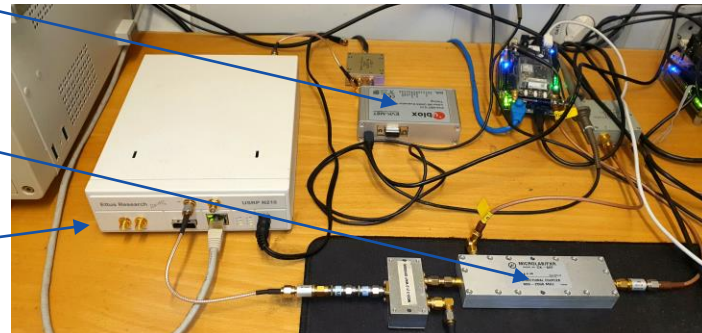
- Attacker receiver used to construct simulated signals identical to the original ones (at first)
 - Carrier/Code phase
 - SNR
 - Doppler
- Synchronized signals
- Lift-off Techniques



Sampling
receiver

Combiner

SDR



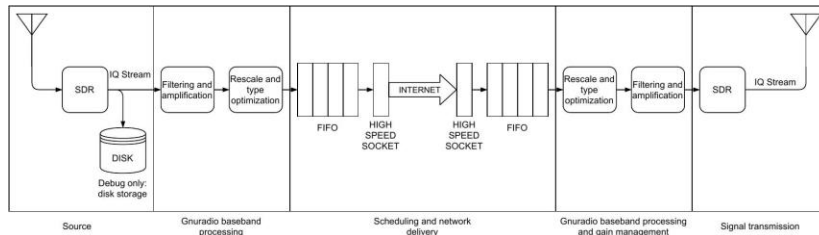
<https://www.ion.org/publications/abstract.cfm?articleID=8132>

Current research: work in progress

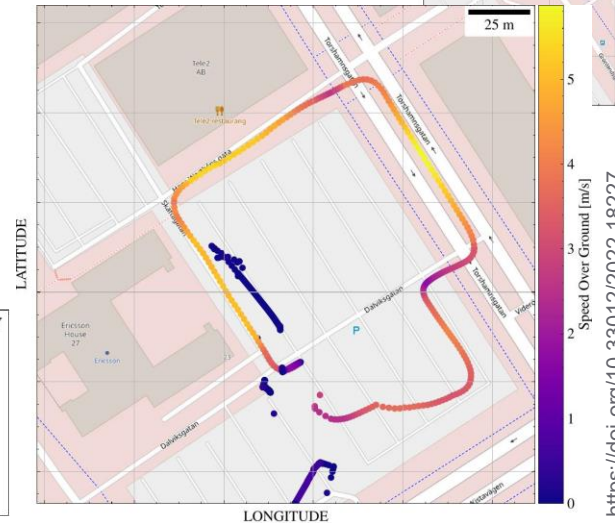
Attacking GNSS Localization (cont'd)

Meaconing

- Meaconing
 - Record/Replay: record in one position and replay in another. What is the difference with simulation then?
 - Relay: record and real-time transmit to a second location for replay
- Works against advanced signals
- Limitation: not flexible, cannot edit signals



Relayed trace



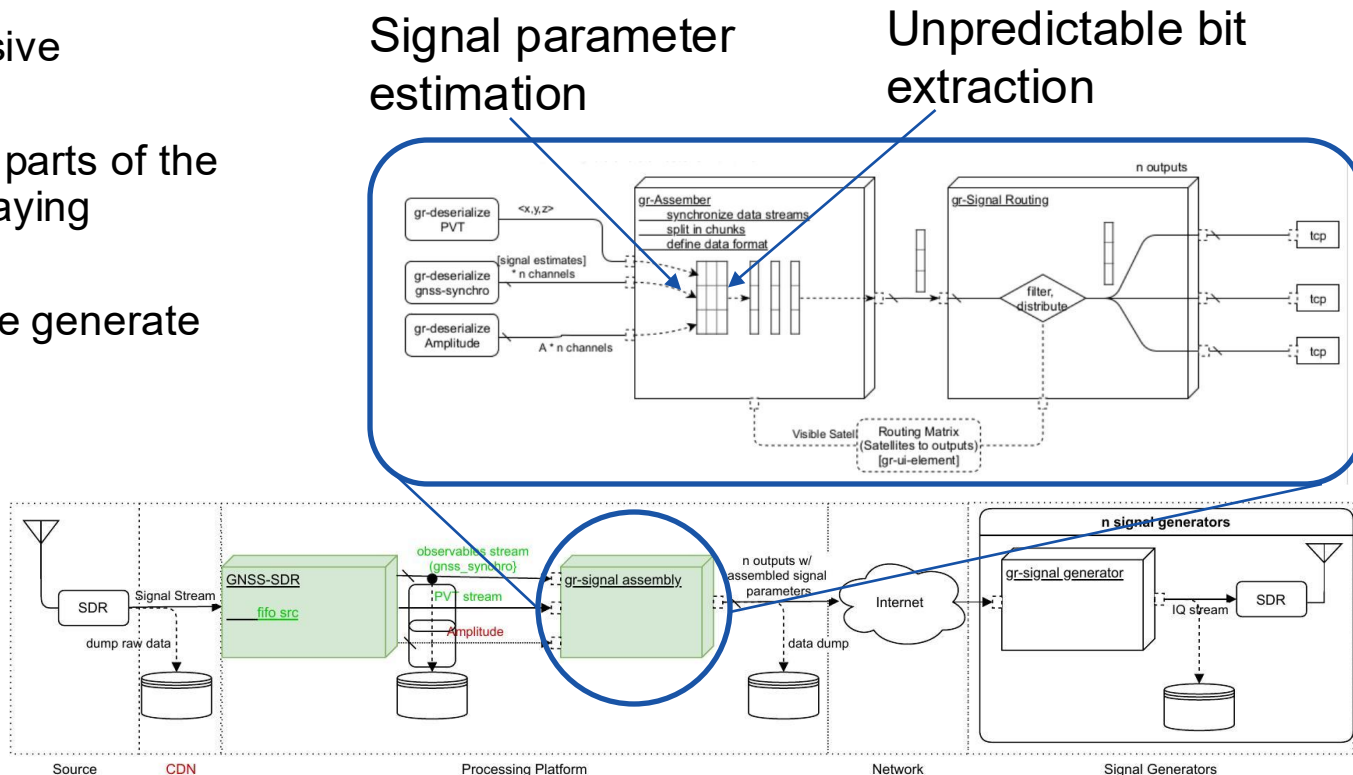
Original trace

<https://doi.org/10.33012/2022.18227>

Attacking GNSS Localization (cont'd)

Meaconing

- Meaconing is expensive bandwidth-wise
- Improvement: not all parts of the signal are worth replaying
- Hint: some parts are unpredictable, can we generate them?



<https://doi.org/10.33012/2022.18227>

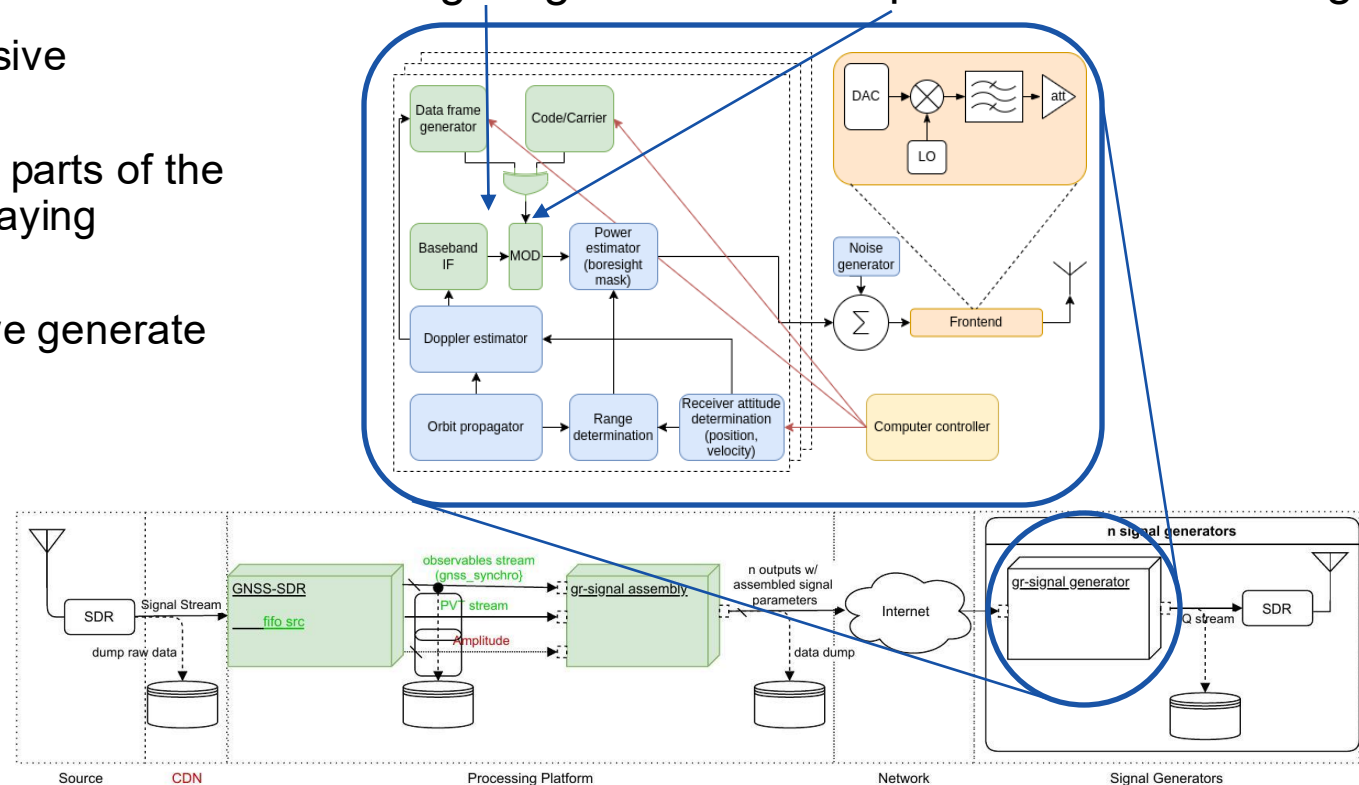
Attacking GNSS Localization (cont'd)

Meaconing

- Meaconing is expensive bandwidth-wise
- Improvement: not all parts of the signal are worth replaying
- Hint: some parts are unpredictable, can we generate them?

Signal generation

Unpredictable bit stuffing

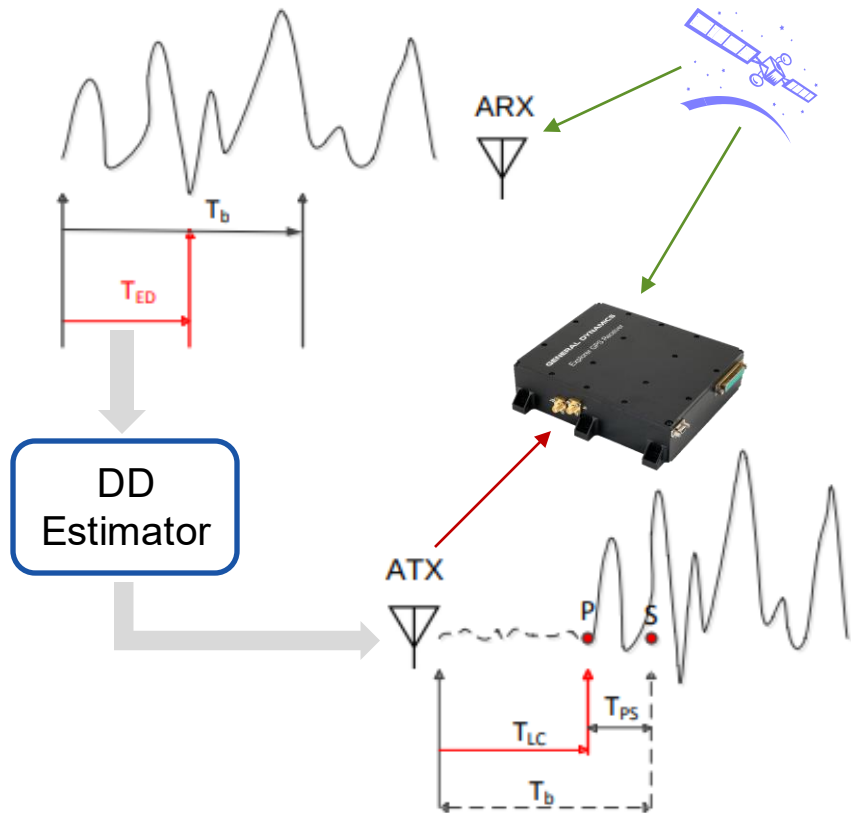
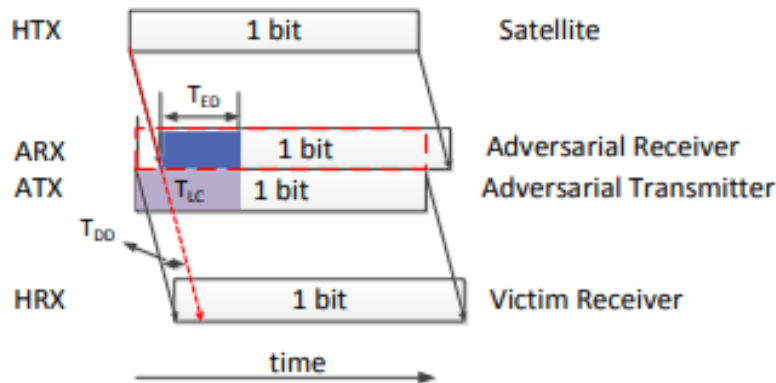


<https://doi.org/10.33012/2022.18227>

Attacking GNSS Localization (cont'd)

Distance decreasing attacks

- DD attacks change the perceived HTX – HRX distance
- GNSS use pseudoranges (practically, distance estimation) to calculate the receiver position
- Mixture of replay and prediction of the symbols





Attacking GNSS Localization (cont'd)

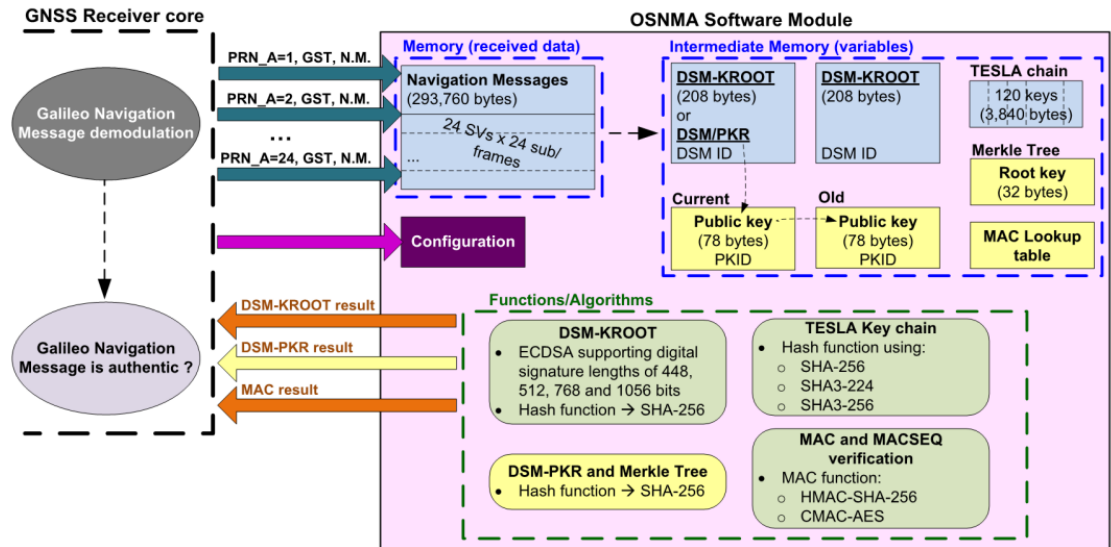
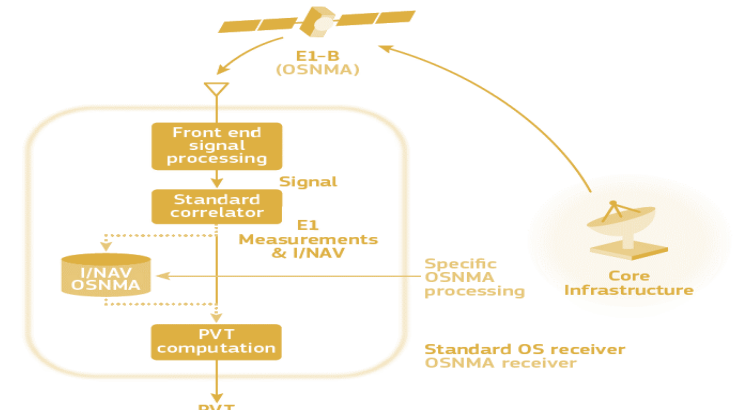
Spoofing: reading list

- P. Papadimitratos and A. Jovanovic, "GNSS-based Positioning: Attacks and countermeasures," *MILCOM 2008 - 2008 IEEE Military Communications Conference*, 2008, pp. 1-7, doi: 10.1109/MILCOM.2008.4753512.
- M. L. Psiaki and T. E. Humphreys, "GNSS Spoofing and Detection," in *Proceedings of the IEEE*, vol. 104, no. 6, pp. 1258-1270, June 2016, doi: 10.1109/JPROC.2016.2526658.
- Humphreys, Todd E., Ledvina, Brent M. et Al., "Assessing the Spoofing Threat: Development of a Portable GPS Civilian Spoofer," *Proceedings of ION GNSS 2008*, Savannah, GA, September 2008, pp. 2314-2325.
- Blum, R., Dötterböck, D., Pany, T., "Investigation of the Vulnerability of Mobile Networks Against Spoofing Attacks on their GNSS Timing-receiver and Developing a Meaconing Protection," *Proceedings of the 2019 ITM*, Reston, Virginia, January 2019, pp. 345-362.
- W. Feng, J. -M. Friedt, G. Goavec-Merou and F. Meyer, "Software-Defined Radio Implemented GPS Spoofing and Its Computationally Efficient Detection and Suppression," in *IEEE Aerospace and Electronic Systems Magazine*, vol. 36, no. 3, pp. 36-52, 1 March 2021, doi: 10.1109/MAES.2020.3040491.
- M. Lenhart, M. Spanghero, P. Papadimitratos, "Distributed and Mobile Message Level Relaying/Replaying of GNSS Signals," *Proceedings of the 2022 International Technical Meeting of The Institute of Navigation*, Long Beach, California, January 2022, pp. 56-67. doi: 10.33012/2022.18227
- Tippenhauer, N. O., Pöpper, C., Rasmussen, K. B., & Čapkun, S. (2011). On the requirements for successful GPS spoofing attacks. *Proceedings of the ACM Conference on Computer and Communications Security*, 75–85. <https://doi.org/10.1145/2046707.2046719>
- K. Zhang and P. Papadimitratos, "On the Effects of Distance-decreasing Attacks on Cryptographically Protected GNSS Signals," *Proceedings of the 2019 International Technical Meeting of The Institute of Navigation*, Reston, Virginia, January 2019, pp. 363-372. <https://doi.org/10.33012/2019.16736>

Defending GNSS (cont'd)

Based on signal structure

- Effective but requires changes to the SiS and to the receiver
- OS-NMA
 - First deployed authenticated civilian GNSS (public testing, release 2023)
 - Constellation exists, but receivers are not available yet (COTS at least)
 - > Giano project: <https://www.ion.org/publications/pdf.cfm?articleID=17714>
 - Anti-replay and anti-spoof features but some attacks are still possible

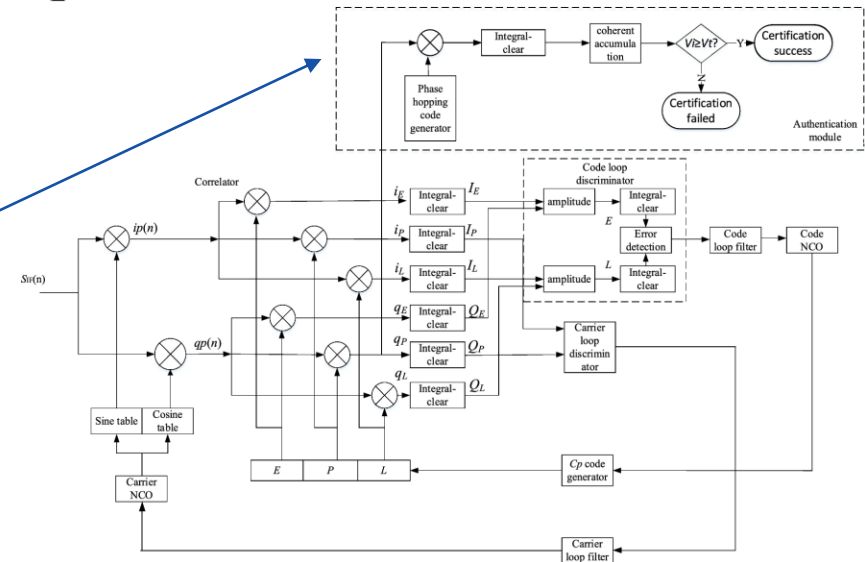
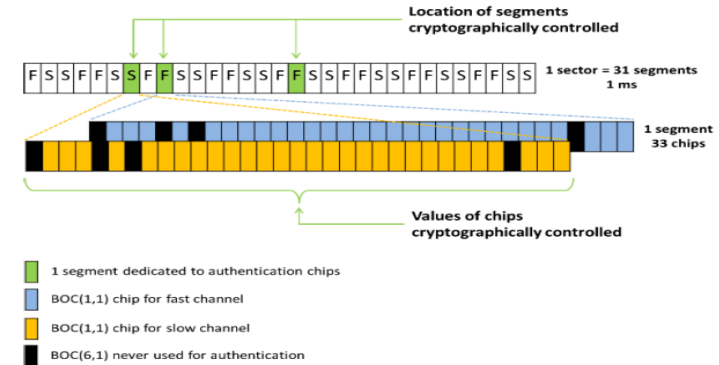


<https://doi.org/10.33012/2020.17714>

Defending GNSS (cont'd)

Based on signal structure

- Effective but requires changes to the SiS and to the receiver
- GPS CHIMERA
 - Chip-message robust authentication
 - Specific to the new generation of GPS satellites
 - Robust anti-spoofing techniques: authentication and watermarking of the signals
- **Drawback:** not backward compatible, thus needs brand new receivers
- Based on Spreading Code Authentication: superior to NMA but adds computational complexity
- Allows individual signal rejection





Defending GNSS (cont'd)

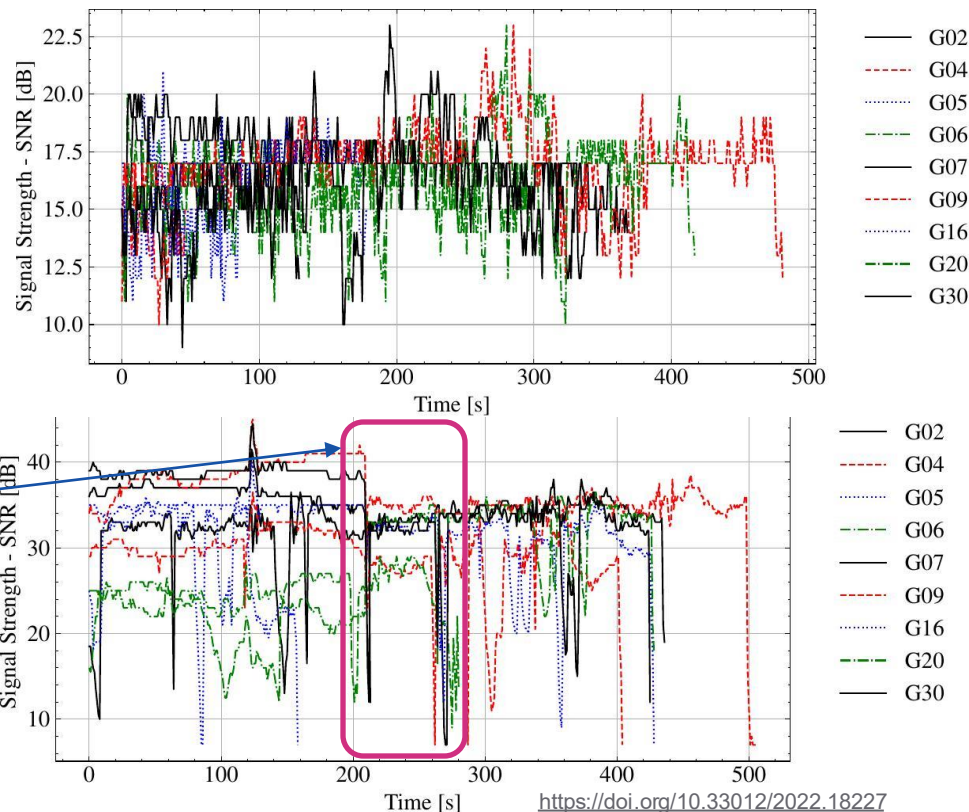
GNSS authentication: reading list

- T. E. Humphreys, "Detection Strategy for Cryptographic GNSS Anti-Spoofing," in *IEEE Transactions on Aerospace and Electronic Systems*, vol. 49, no. 2, pp. 1073-1090, APRIL 2013, doi: 10.1109/TAES.2013.6494400.
- F. Gallardo and A. P. Yuste, "SCER Spoofing Attacks on the Galileo Open Service and Machine Learning Techniques for End-User Protection," in *IEEE Access*, vol. 8, pp. 85515-85532, 2020, doi: 10.1109/ACCESS.2020.2992119.
- Fernández-Hernández, I., Rijmen, V., Seco-Granados, G., Simon, J., Rodríguez, I., and Calle, J. D. (Spring) A Navigation Message Authentication Proposal for the Galileo Open Service. *J Inst Navig*, 63: 85– 102. doi: [10.1002/navi.125](https://doi.org/10.1002/navi.125).
- Anderson, Jon M., Carroll, Katherine L., et Al., "Chips-Message Robust Authentication (Chimera) for GPS Civilian Signals," Proceedings of the 30th International Technical Meeting of the Satellite Division of The Institute of Navigation (ION GNSS+ 2017), Portland, Oregon, September 2017, pp. 2388-2416. doi: [10.33012/2017.15206](https://doi.org/10.33012/2017.15206)
- K. Zhang, E. G. Larsson and P. Papadimitratos, "Protecting GNSS Open Service Navigation Message Authentication Against Distance-Decreasing Attacks," in *IEEE Transactions on Aerospace and Electronic Systems*, vol. 58, no. 2, pp. 1224-1240, April 2022, doi: 10.1109/TAES.2021.3122512.
- Wesson, K., Rothlisberger, M. and Humphreys, T. (2012), Practical Cryptographic Civil GPS Signal Authentication. *J Inst Navig*, 59: 177-193. <https://doi.org/10.1002/navi.14>
- Arizabaleta, Markel, Gkougkas, Elias, Pany, Thomas, "A Feasibility Study and Risk Assessment of Security Code Estimation and Replay (SCER) Attacks," *Proceedings of ION GNSS+ 2019*, Miami, Florida, September 2019, pp.1039-1050. <https://doi.org/10.33012/2019.17041>

Defending GNSS (cont'd)

Based on signal properties

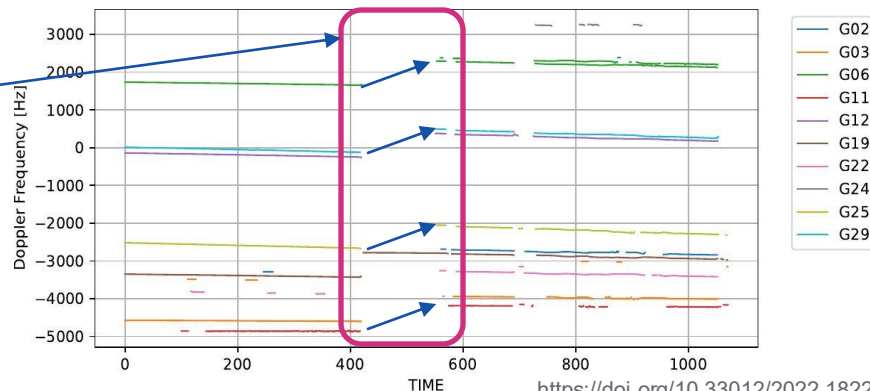
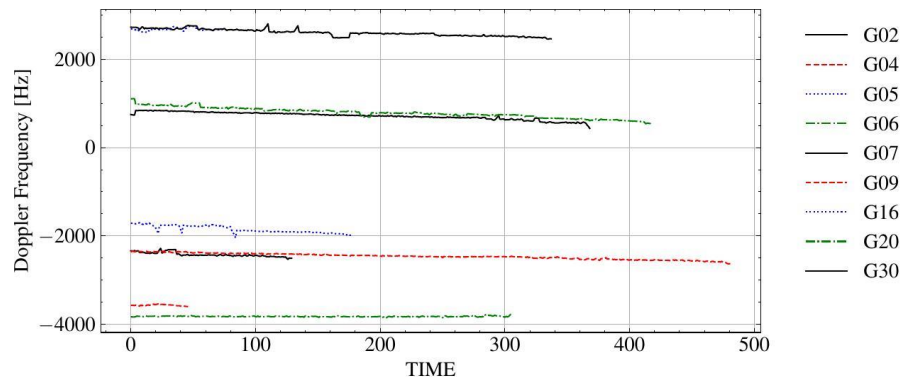
- Effective if the receiver provides low level measurements of the signals (before PVT)
- Carrier phase/ CN_0 are powerful tools to detect changes in the signal, but are also dependent on the environment
- Drop in signal strength at the beginning of the replay (not power matched)
- Requires calibrated frontend and knowledge of the IMP propagation of the signal



Defending GNSS (cont'd)

Based on signal properties

- Effective if the receiver provides low level measurements of the signals (before PVT)
- Carrier phase/ CN_0 to detect changes in the signal: dependent on the environment
- Doppler skip due to the replay: the transmitter is not synchronized with the real signals and produces unrealistic doppler
- Requires high quality clocks at the sampling/transmitting radio



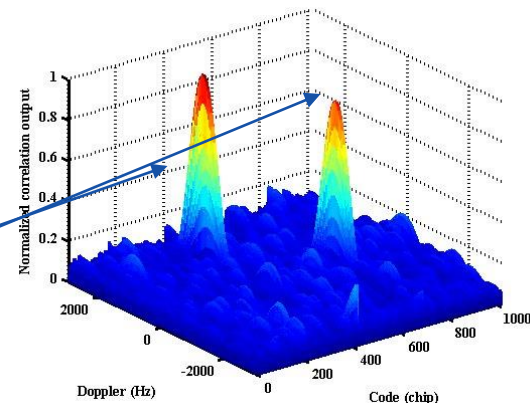
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Defending GNSS (cont'd)

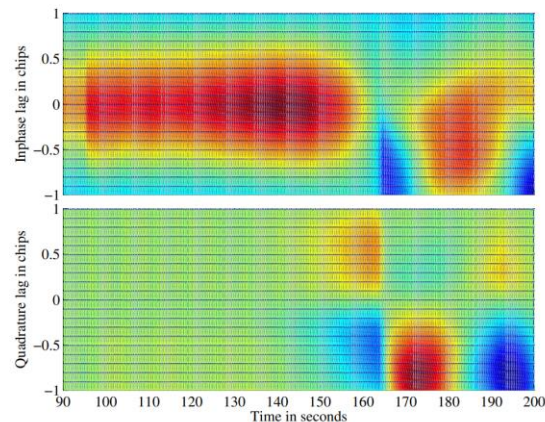
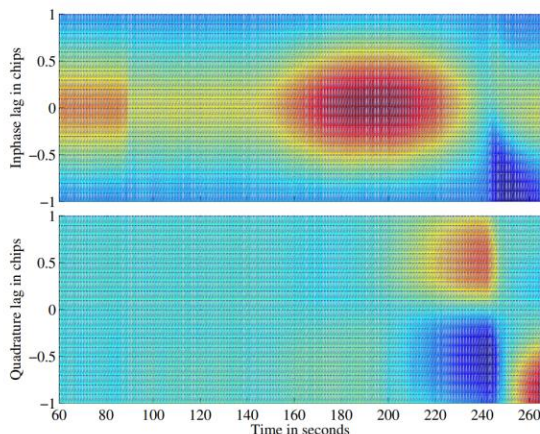
Based on signal properties

- Acquisition peak tracking and IQ imbalance show the presence of multiple signals interacting with each other (both if synchronous and asynchronous)
- Great resource:
<https://developer.android.com/guide/topics/sensors/gnss>

Clear spoofing-induced peak splitting: why can't we have more than one peak?



<http://dx.doi.org/10.1109/PLANS.2012.6236917>



https://radionavlab.ae.utexas.edu/images/stories/files/papers/tbION_for_distribution.pdf



Attacking GNSS (cont'd)

Spoofing detection based on signal properties: reading list

- Nicholas Spens, Dong-Kyeong Lee, Filip Nedelkov, and Dennis Akos, "Detecting GNSS Jamming and Spoofing on Android Devices", *NAVIGATION: Journal of the Institute of Navigation* September 2022, 69 (3) navi.537; DOI: <https://doi.org/10.33012/navi.537>
- N. Stenberg, E. Axell, J. Rantakokko and G. Hendeby, "GNSS Spoofing Mitigation Using Multiple Receivers," *2020 IEEE/ION Position, Location and Navigation Symposium (PLANS)*, 2020, pp. 555-565, doi: 10.1109/PLANS46316.2020.9109958.
- Psiaki, M.L.; O'Hanlon, B.W.; Powell, S.P.; Bhatti, J.A.; Humphreys, T.E.; Schofield, A. GNSS lies, GNSS truth: Spoofing detection with two-antenna differential carrier phase. *GPS World* 2014, 25, 36–44
- Humphreys, T.E.; Bhatti, J.A.; Shepard, D.P.; Wesson, K.D. The Texas Spoofing Test Battery: Toward a Standard for Evaluating GPS Signal Authentication Techniques. In *Phenomena in Ionized Gases*, IEEE, VI International Conference, Volume III; Nashville Convention Center: Nashville, TN, USA, 2012.
- Akos, D.M. (2012), Who's Afraid of the Spoofer? GPS/GNSS Spoofing Detection via Automatic Gain Control (AGC). *J Inst Navig*, 59: 281-290. <https://doi.org/10.1002/navi.19>
- Psiaki, M.L., Powell, S.P., O'Hanlon, B.W., "GNSS Spoofing Detection using High-Frequency Antenna Motion and Carrier-Phase Data," *Proceedings of the 26th International Technical Meeting of the Satellite Division of The Institute of Navigation (ION GNSS+ 2013)*, Nashville, TN, September 2013, pp. 2949-2991.
- Harshad Sathaye and Aanjan Ranganathan. 2020. *SemperFi: a spoofer eliminating standalone GPS receiver*. In *Proceedings of the 13th ACM Conference on Security and Privacy in Wireless and Mobile Networks (WiSec '20)*. Association for Computing Machinery, New York, NY, USA, 353–355. <https://doi.org/10.1145/3395351.3401703>
- E. Schmidt, J. Lee, N. Gatsis and D. Akopian, "Rejection of Smooth GPS Time Synchronization Attacks via Sparse Techniques," in *IEEE Sensors Journal*, vol. 21, no. 1, pp. 776-789, 1 Jan.1, 2021, doi: 10.1109/JSEN.2020.3014239.

Defending GNSS (cont'd)

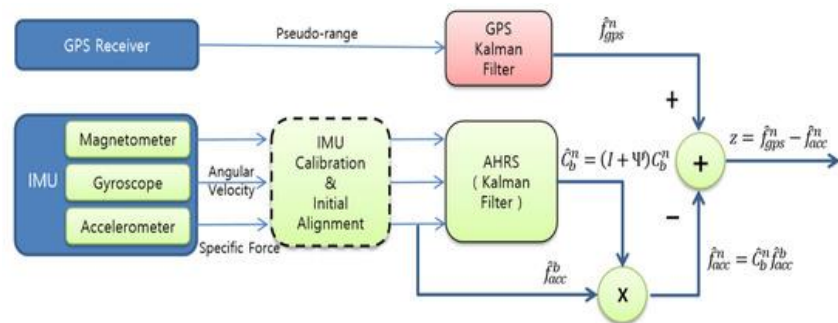
Based on other sensors

- GNSS receivers are often integrate in sensor-rich, mobile, connected devices.
- Exploit onboard resources to detect PVT inconsistency
 - AHRs (attitude heading reference system)
 - Clocks
 - Signals of Opportunity
 - Other constellations
- Data-level, computationally effective technique
- Requires additional hardware



<https://www.oscilloquartz.com/en>

<https://www.oriola.com/jamming-and-spoofing-protection-for-uavs-using-gnss-signals/>



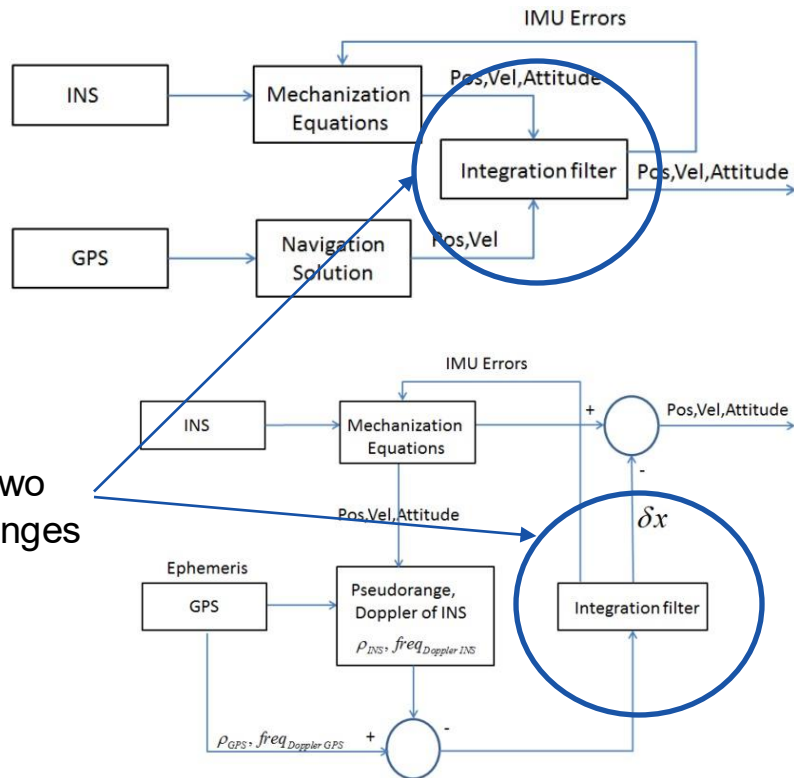
<https://doi.org/10.3390/s20040954>

Defending GNSS (cont'd)

Based on other sensors - INS/AHRS

- GNSS coupling with INS/AHRS allows to measure the displacement and the orientation of the GNSS receiver
- In normal case: INS can be used to improve the GNSS solution
 - Loose coupling: data-level filtering and PV fusion
 - Tight coupling: GNSS raw measurements (doppler and pseudorange variations) are co-estimated with the INS
- Anti-spoofing is intuitive: how?

Compare the two solutions / changes if consistent!





Defending GNSS (cont'd)

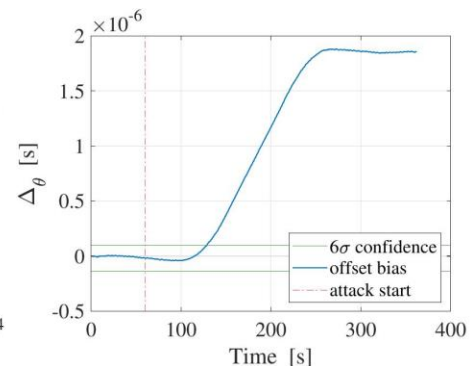
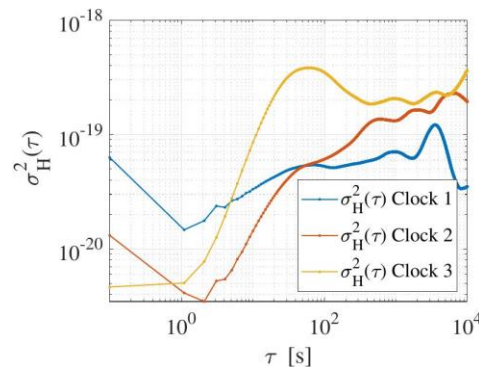
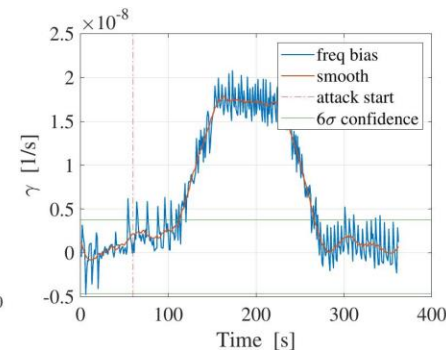
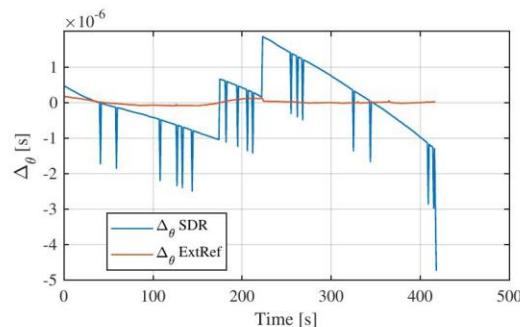
Spoofing detection based on INS/AHRS: reading list

- Broumandan, A.; Lachapelle, G. Spoofing Detection Using GNSS/INS/Odometer Coupling for Vehicular Navigation. Sensors 2018, 18, 1305. <https://doi.org/10.3390/s18051305>
- Kwon, K.-C.; Shim, D.-S. Performance Analysis of Direct GPS Spoofing Detection Method with AHRS/Accelerometer. Sensors 2020, 20, 954. <https://doi.org/10.3390/s20040954>
- Gao, Y.; Li, G. A Slowly Varying Spoofing Algorithm on Loosely Coupled GNSS/IMU Avoiding Multiple Anti-Spoofing Techniques. Sensors 2022, 22, 4503. <https://doi.org/10.3390/s22124503>
- Curran, James T. and Ali Broumandan. "On the use of Low-Cost IMUs for GNSS Spoofing Detection in Vehicular Applications." (2017).
- Clements, Zachary, Yoder, James E., Humphreys, Todd E., "Carrier-phase and IMU based GNSS Spoofing Detection for Ground Vehicles," Proceedings of the 2022 International Technical Meeting of The Institute of Navigation, Long Beach, California, January 2022, pp. 83-95. <https://doi.org/10.33012/2022.18252>

Defending GNSS (cont'd)

Based on other sensors - clocks

- GNSS precisely measures time offsets
 - We lock the receiver clock to the satellites/constellations clock
 - The remote clock is not drifting
- An attacker can target the PV or **T(ime)**
- How to protect the receiver then?
 - Clock stability estimates; not trivial!
 - Need a "good enough" clock at the receiver
 - Extremely powerful! -> not only data, but also signal stability properties can be explored (e.g., frequency offset, dopplers, ...)



10.1109/AERO53065.2022.9843381



Defending GNSS (cont'd)

Spoofing detection based on clocks: reading list

- Hwang, Patrick Y. C. and Gary A. McGraw. "Receiver Autonomous Signal Authentication (RASA) based on clock stability analysis." *2014 IEEE/ION Position, Location and Navigation Symposium - PLANS 2014* (2014): 270-281.
- Md Tanvir Arafin, Dhananjay Anand, and Gang Qu. 2017. *A Low-Cost GPS Spoofing Detector Design for Internet of Things (IoT) Applications. In Proceedings of the on Great Lakes Symposium on VLSI 2017 (GLSVLSI '17). Association for Computing Machinery, New York, NY, USA, 161–166. <https://doi.org/10.1145/3060403.3060455>*
- K. Zhang, M. Spanghero and P. Papadimitratos, "Protecting GNSS-based Services using Time Offset Validation," *2020 IEEE/ION Position, Location and Navigation Symposium (PLANS)*, 2020, pp. 575-583, doi: 10.1109/PLANS46316.2020.9110224.
- M. Spanghero and P. Papadimitratos, "High-precision Hardware Oscillators Ensemble for GNSS Attack Detection," *2022 IEEE Aerospace Conference (AERO)*, 2022, pp. 1-11, doi: 10.1109/AERO53065.2022.9843381.
- Marnach, D., Mauw, S., Martins, M., & Harpes, C. (2013). Detecting Meaconing Attacks by Analysing the Clock Bias of GNSS Receivers. *Artificial Satellites*, 48(2), 63–83. <https://doi.org/10.2478/arsa-2013-0006>
- Gaggero, P. O., & Borio, D. (2008). Ultra-stable oscillators: Limits of GNSS coherent integration. *21st International Technical Meeting of the Satellite Division of the Institute of Navigation, ION GNSS 2008*, 1, 101–111.
- Jafarnia-Jahromi, A., Daneshmand, S., Broumandan, A., Nielsen, J., & Lachapelle, G. (2013). PVT solution authentication based on monitoring the clock state for a moving GNSS receiver. *European Navigation Conference (ENC)*, 11.

Defending GNSS (cont'd)

Based on signals of opportunity

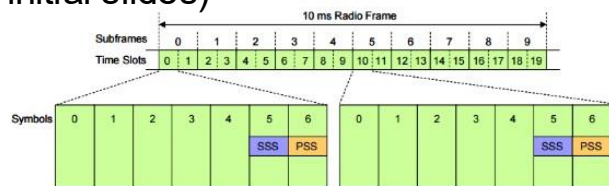
- Not only GNSS signals can provide navigation:
 - Given the right hardware
 - Given enough stability
 - Any repeating pattern can be used for range estimation and localization (remember initial slides)

OFDM Transmitters

- Like TV towers

LTE signal

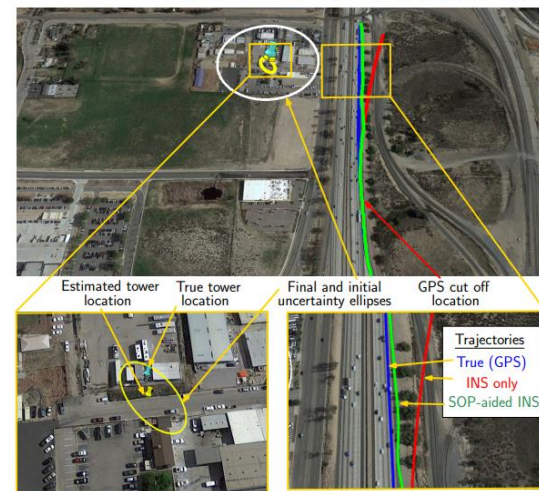
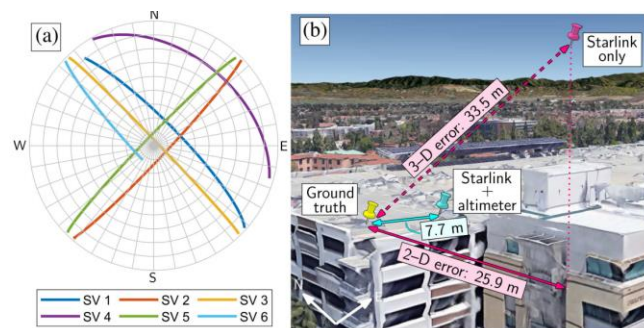
- PSS** and **SSS** signals used to synchronize UE and RAN



<https://www.techplayon.com/pss-and-sss-lte-synchronization/>

Opportunistic LEO signals

- Starlink** : <https://www.technologyreview.com/2022/10/21/1062001/spacex-starlink-signals-reverse-engineered-gps/>





Defending GNSS (cont'd)

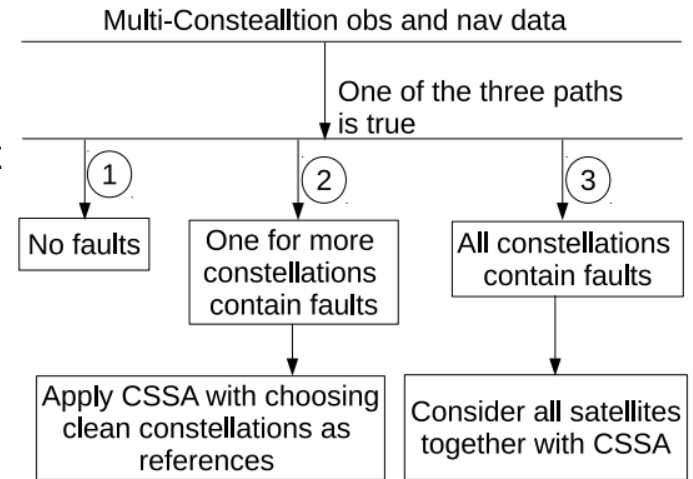
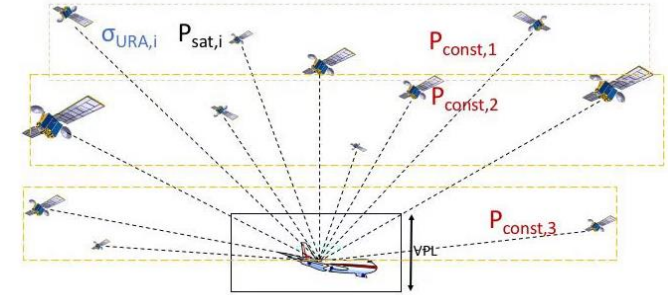
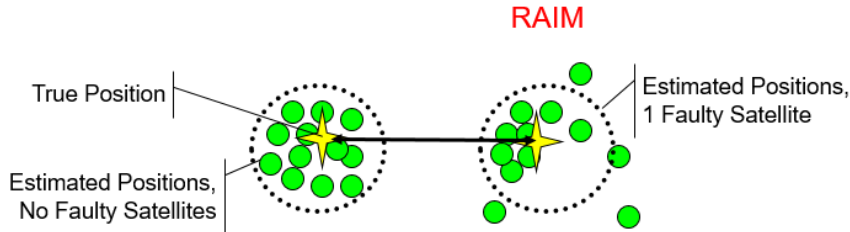
Based on signals of opportunity: reading list

- Z. M. Kassas, J. Khalife, A. A. Abdallah and C. Lee, "I Am Not Afraid of the GPS Jammer: Resilient Navigation Via Signals of Opportunity in GPS-Denied Environments," in *IEEE Aerospace and Electronic Systems Magazine*, vol. 37, no. 7, pp. 4-19, 1 July 2022, doi: 10.1109/MAES.2022.3154110.
- Mark L. Psiaki, Brian D. Slosman, "Tracking Digital FM OFDM Signals for the Determination of Navigation Observables", *NAVIGATION: Journal of the Institute of Navigation* Jun 2022, 69 (2) navi.521; DOI: 10.33012/navi.521
- Neinavaie, Mohammad, Khalife, Joe, Kassas, Zaher M., "Exploiting Starlink Signals for Navigation: First Results," *Proceedings of the 34th International Technical Meeting of the Satellite Division of The Institute of Navigation (ION GNSS+ 2021)*, St. Louis, Missouri, September 2021, pp. 2766-2773. <https://doi.org/10.33012/2021.18122>
- Perdue, Lisa, Fischer, John, Dries, Ronald, "Signals of Opportunity as an Augmentation or Alternative to GNSS for Critical Timing Applications," *Proceedings of the 48th Annual Precise Time and Time Interval Systems and Applications Meeting*, Monterey, California, January 2017, pp. 171-176. <https://doi.org/10.33012/2017.14988>
- Z. M. Kassas, "Navigation with Cellular Signals of Opportunity," in *Position, Navigation, and Timing Technologies in the 21st Century: Integrated Satellite Navigation, Sensor Systems, and Civil Applications*, IEEE, 2021, pp.1171-1223, doi: 10.1002/9781119458555.ch38.

Defending GNSS (cont'd)

Based on multiple GNSS constellations

- Single GNSS constellation only need 4 satellites for a fully resolved PNT
 - What if we have more satellites?
 - What if we have more constellations available?
- Calculate different PNTs with different subsets: RAIM
- Integrate multiple constellations for better accuracy
- Cluster solutions from different constellations to detect attackers: Cluster based solution separation (CSSA)





Defending GNSS (cont'd)

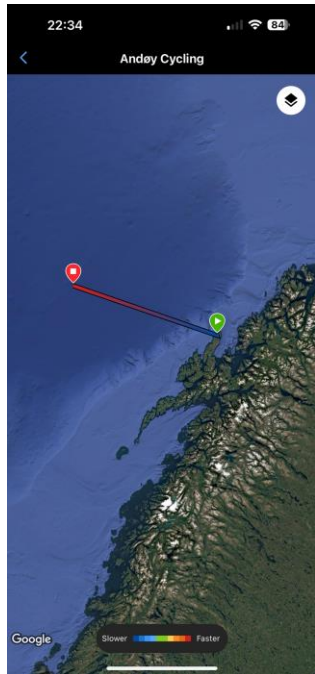
Based on multiple GNSS constellations: reading list

- Granat R, Donnellan A, Heflin M, Lyzenga G, Glasscoe M, Parker J, Pierce M, Wang J, Rundle J, Ludwig LG. Clustering Analysis Methods for GNSS Observations: A Data-Driven Approach to Identifying California's Major Faults. *Earth Space Sci.* 2021 Nov;8(11):e2021EA001680. doi: 10.1029/2021EA001680. Epub 2021 Oct 29. PMID: 34820480; PMCID: PMC8596415.
- K. Zhang and P. Papadimitratos, "Secure Multi-Constellation GNSS Receivers with Clustering-Based Solution Separation Algorithm," *2019 IEEE Aerospace Conference*, 2019, pp. 1-9, doi: 10.1109/AERO.2019.8742021.
- K. Zhang and P. Papadimitratos, "Fast Multiple Fault Detection and Exclusion (FM-FDE) Algorithm for Standalone GNSS Receivers," in *IEEE Open Journal of the Communications Society*, vol. 2, pp. 217-234, 2021, doi: 10.1109/OJCOMS.2021.3050333.
- J. Blanch, T. Walter and P. Enge, "RAIM with Optimal Integrity and Continuity Allocations Under Multiple Failures," in *IEEE Transactions on Aerospace and Electronic Systems*, vol. 46, no. 3, pp. 1235-1247, July 2010, doi: 10.1109/TAES.2010.5545186.
- Blanch, Juan, Walter, Todd, Enge, Per, Lee, Young, Pervan, Boris, Rippl, Markus, Spletter, Alex, "Advanced RAIM user Algorithm Description: Integrity Support Message Processing, Fault Detection, Exclusion, and Protection Level Calculation," *Proceedings of the 25th International Technical Meeting of the Satellite Division of The Institute of Navigation (ION GNSS 2012)*, Nashville, TN, September 2012, pp. 2828-2849.



Defending GNSS (cont'd)

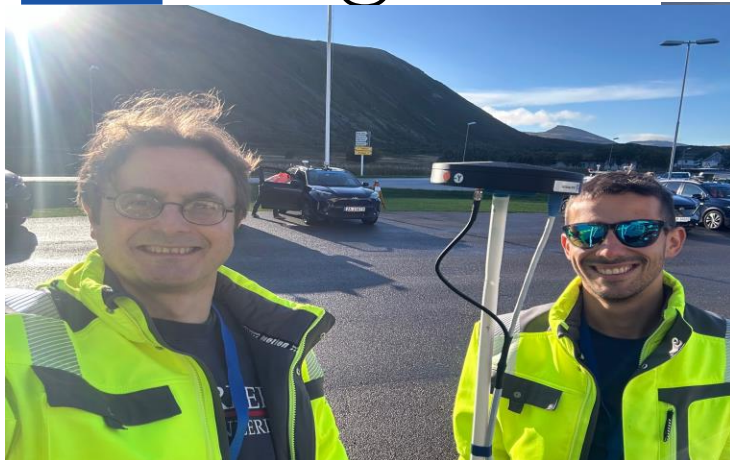
NSS @ Jammertest





Defending GNSS (cont'd)

NSS @ Jammertest





Autumn 2025

Discussion leads

Andrew Roberts &
Kostas
Kalogiannis

Secure localization and positioning

Panos Papadimitratos

www.eecs.kth.se/nss

